

Probability

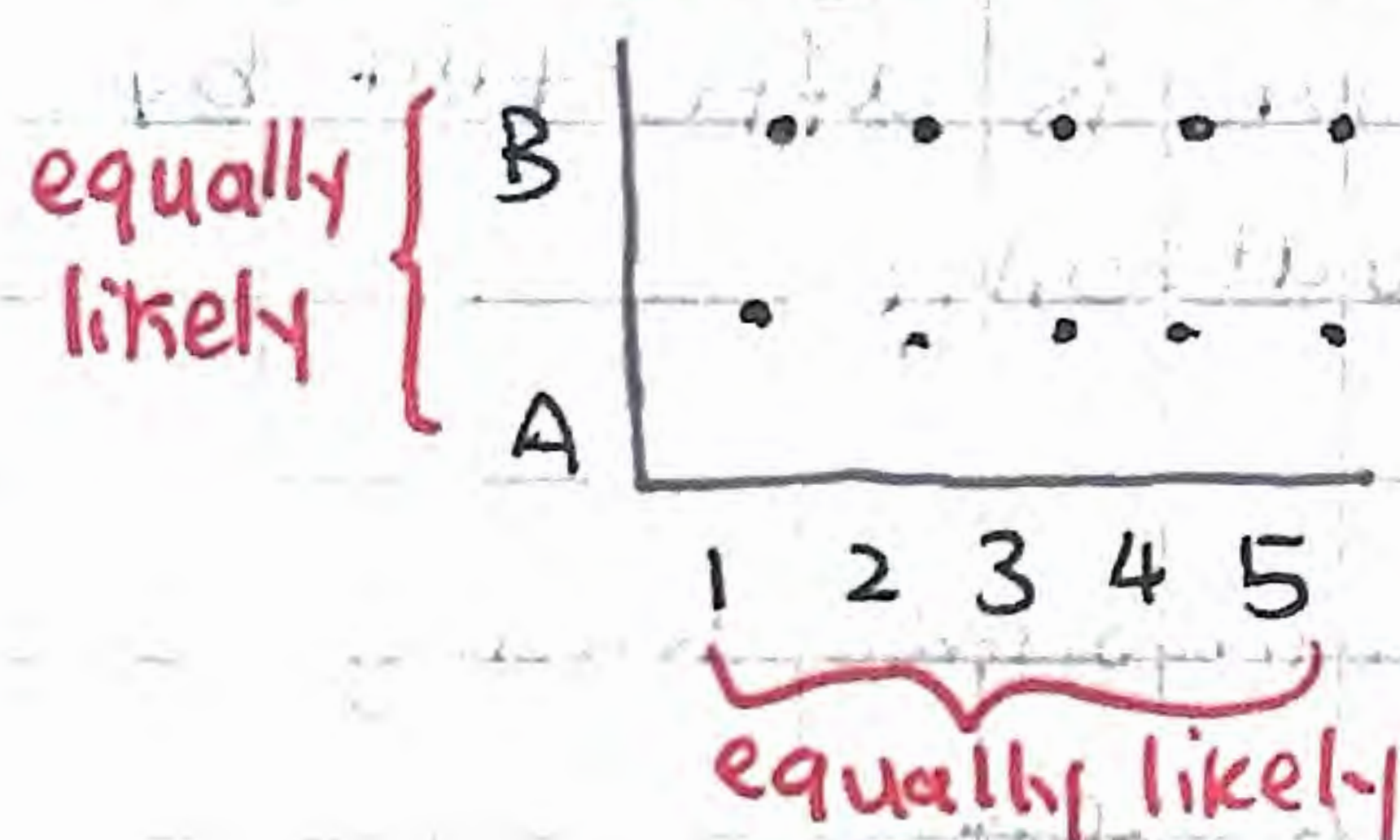
Notation: $P(A) + P(A') = 1$ $P(A)$ refers to the probability event A happens
 $P(A')$ refers to the probability event A doesn't happen
 $P(A \cup B)$ refers to event A or B happening (union)
 $P(A \cap B)$ refers to event A and B happening (intersection)

Sample space - Set of all possible outcomes of an experiment. (total)

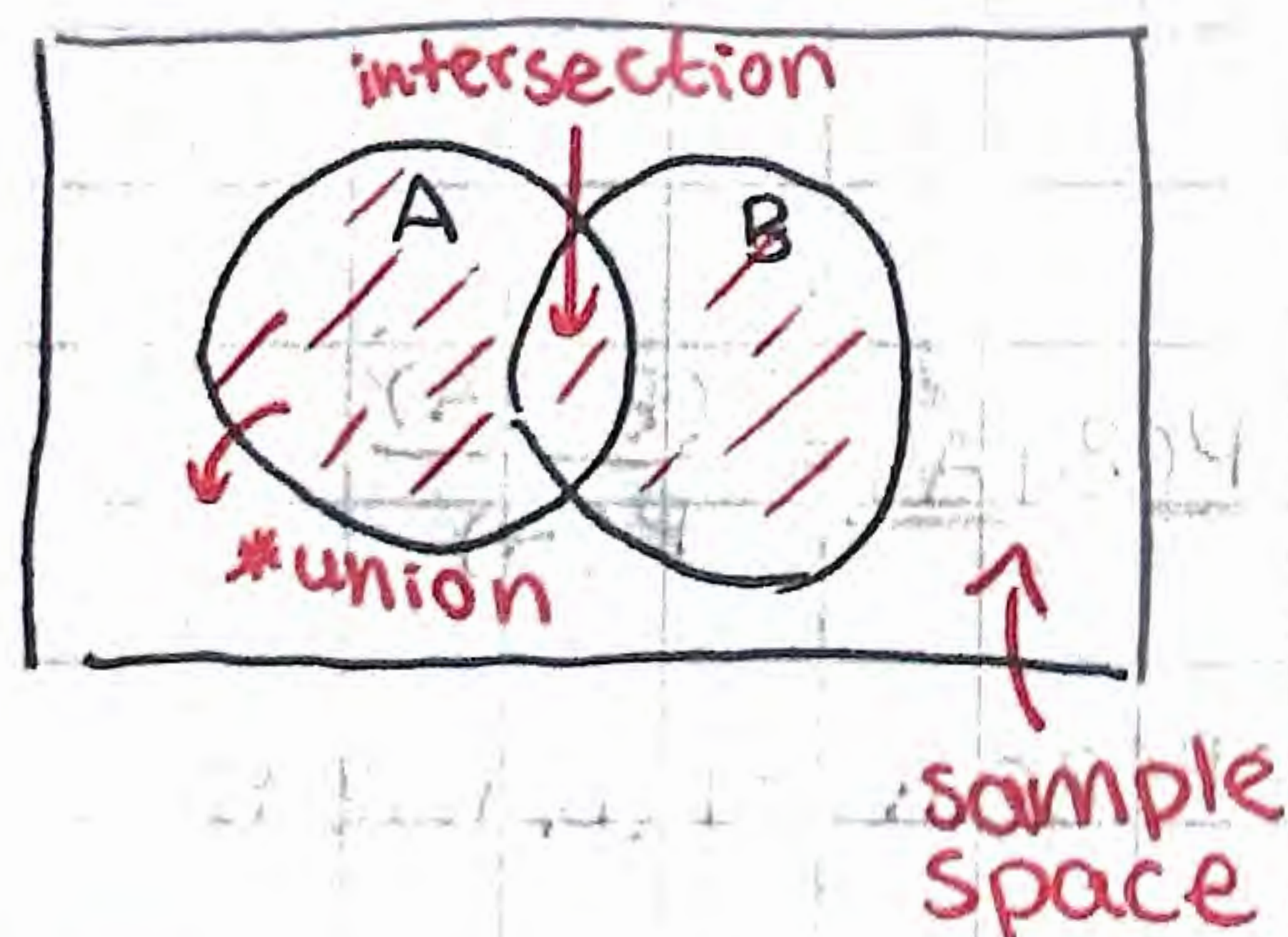
Two events are considered complimentary when there's always exactly one of the two events happening

There are 3 main ways to visualize probability:

1. 2D grid (equally likely event outcomes)



Each point represents unique combined outcome in the sample space

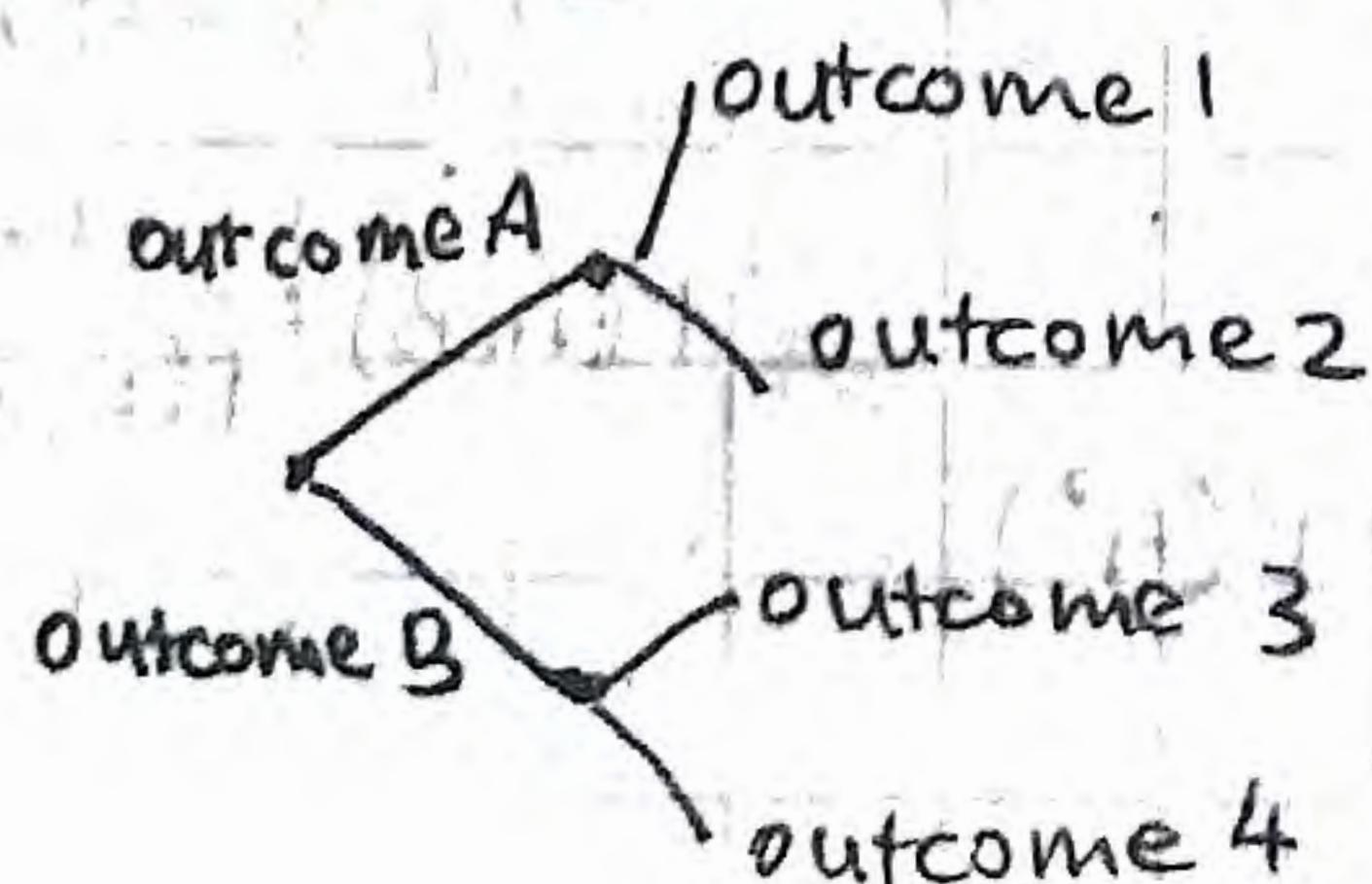


2. Venn Diagram

(two/three events and their union/intersect)

Helps visualize probability by area (sum of areas as sum of probabilities)

3. Tree diagram (recursion)



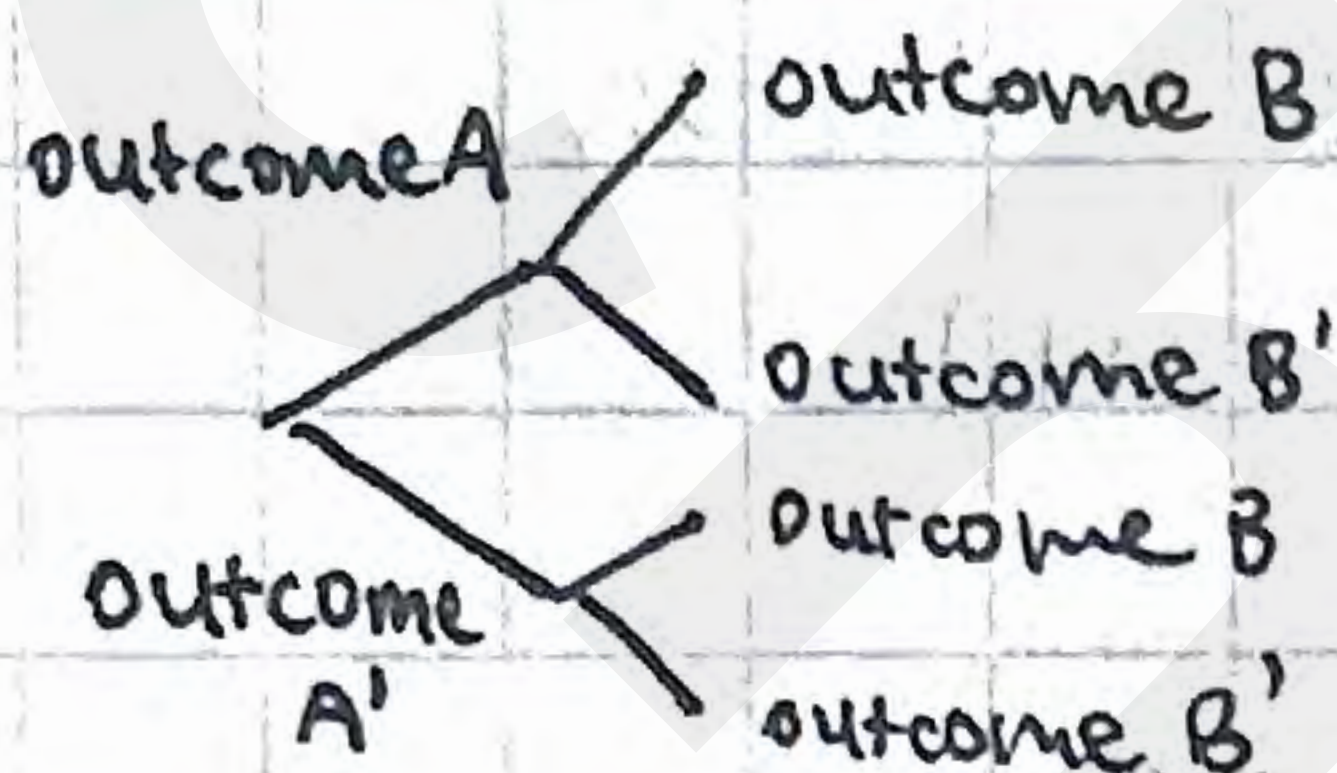
Helps visualize recursive relations

Also probability as a sequence (not happening all at once)

Mutually exclusive - Two events cannot both happen together at once

$$P(A \cap B) = 0 \quad \text{i.e. no intersection between the two}$$

Independent events - An event's outcome does not affect the other

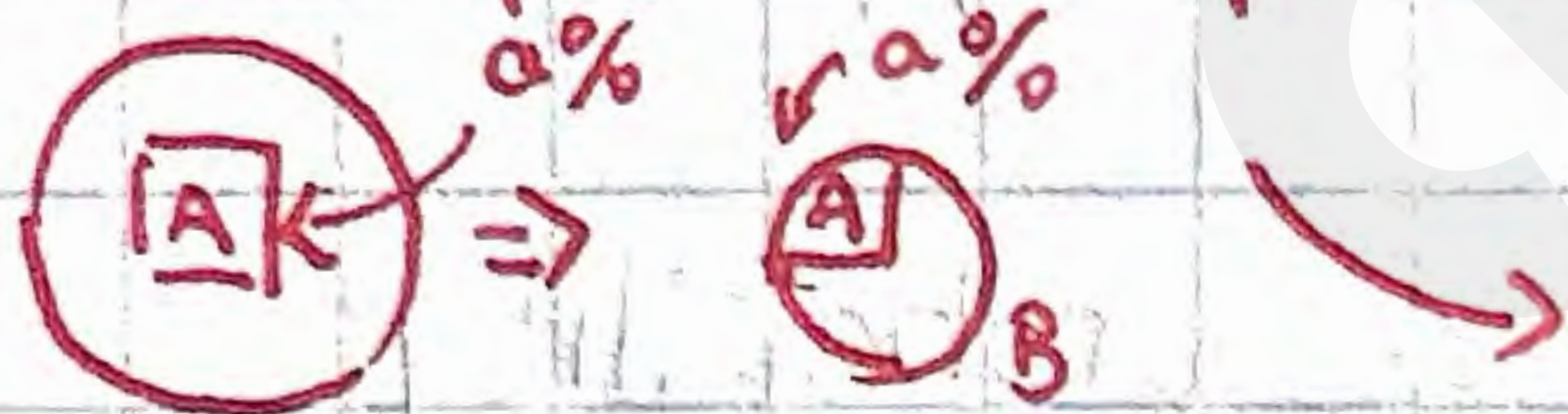


in other words

$$A \text{ given } B = A$$

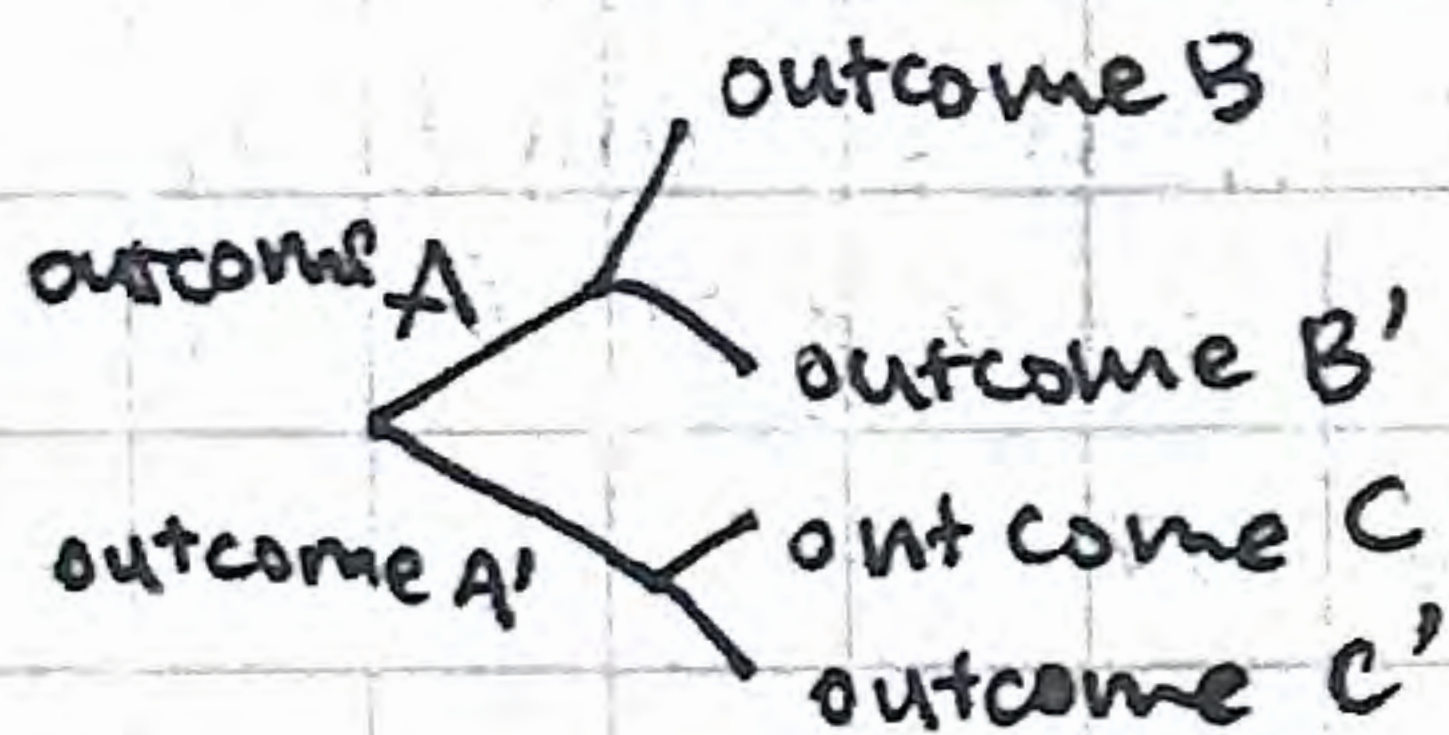
$$P(A|B) = P(A)$$

think of it as a sample size



whether B happens or not, the chance of A happening will be the same

Dependent events - An event's outcome is affected by another event's outcome



Typically in scenarios like taking marbles out of a sack

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{or} \quad P(B|A) = \frac{P(B \cap A)}{P(A)}$$



	B	B'
A		
A'		

Baye's theorem

$$\therefore P(B \cap A) = P(B|A)P(A)$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(B) = P(B \cap A) + P(B \cap A') \\ = P(B|A)P(A) + P(B|A')P(A')$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A')P(A')}$$

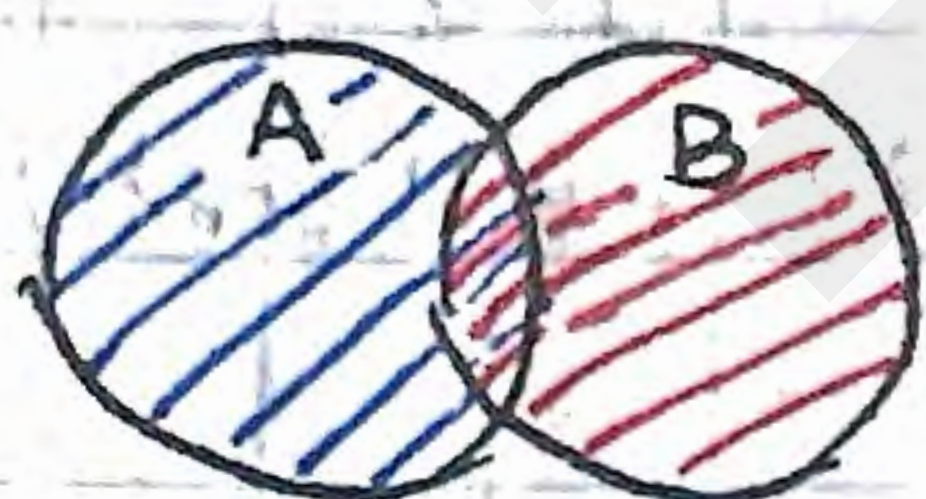
Principle of Inclusion Exclusion

How do you find the intersection or union of events?

You can use multiplication, addition, subtraction

$$P(A \cap B) = P(A) \cdot P(B) \quad \text{intersection}$$

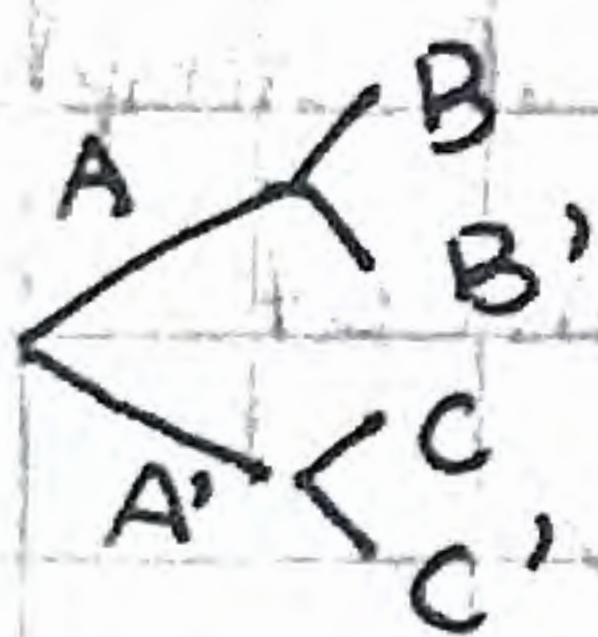
$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad \text{union}$$



This P.I.E. also applies for 3 events, etc.

$$\begin{aligned} P(A \cup B \cup C \text{ etc.}) &= P(A) + P(B) + \dots \\ &\quad - P(A \cap B) - P(B \cap C) - \dots \\ &\quad + P(A \cap B \cap C) + \dots \\ &= P(A) + P(B) + \dots \\ &\quad - P(A)P(B) - P(B)P(C) - \dots \\ &\quad + P(A)P(B)P(C) + \dots \end{aligned}$$

Probability in tree diagrams



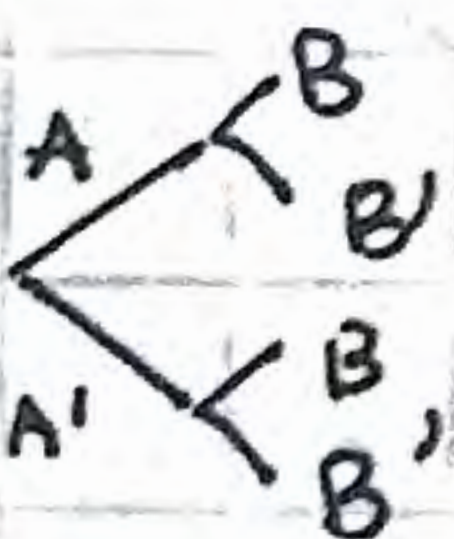
dep.

$$P(B) = a\% \cdot b\%$$

$$P(B|A) = b\%$$

this one already means A will happen, so ignore a%

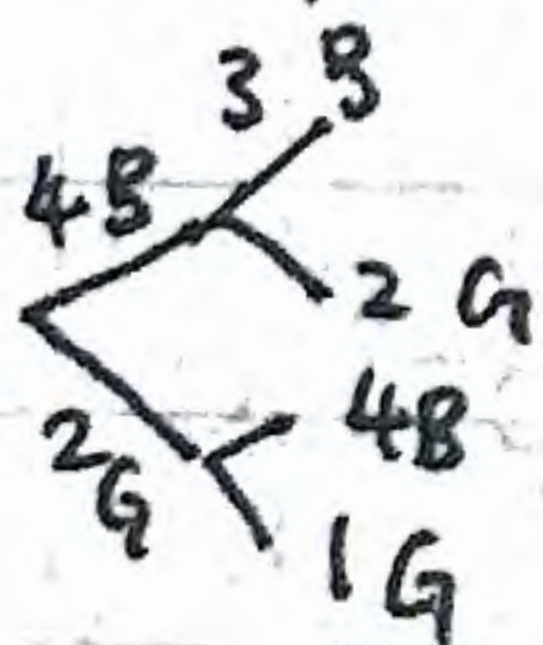
can be used with Baye's theorem (logical approach to Baye's theorem)



indep.

$$P(B) = a\% \cdot b\% + a'\% \cdot b\%$$

E.g. 4B, 2G, one is randomly drawn (no replacement)



$$a) \text{ 2nd is B} = \frac{3}{5} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{4}{5} = \frac{2}{3}$$

$$b) \text{ 1st is G, given 2nd is B} = \frac{4}{5} \cdot \frac{3}{2} = \frac{2}{5}$$

Descriptive Statistics

Sampling is hard to guarantee 100% fairness

There are different types of sample techniques

1. Simple Random - Simple random selection from the population

• Complete randomness to choose

2. Stratified - Splitting population into stratas, taking same % of randomly selected individuals from each strata

• E.g. 20% of each grade, randomly selected

3. Convenience - Most convenient way of finding samples

• E.g. Sample from people you meet

4. Quota - Similar to stratified but lacks randomness by doing it in the most convenient way of sampling

• E.g. 20% of each grade, selected from people you meet

5. Systematic - Sampling according to some kind of system

• E.g. Every 5th name

Quantitative data - Data measured in numbers used to measure performance and identify trends through statistical analysis

Qualitative data - Non-numerical, descriptive data based on interpretations through language

Mean - The "average": $\frac{\sum x_i}{n}$ $\leftarrow$ sum of all terms
 $\leftarrow$ n terms total

Median - The middlemost term (or the mean of the two middle terms)

Mode - The term of greatest frequency

Range - Difference between the maximum and minimum terms

Standard Deviation - Measure of how spread out a set of values is relative to the mean
 σ (or $\sqrt{\text{variance}}$)

Variance - Measure of the dispersion of a set of data around its average
 σ^2

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$\leftarrow$ ith term $\leftarrow$ mean
 $\leftarrow$ n total terms

* You can quickly calculate all this with calculator
[F12] spreadsheet (enter data by rows)
Menu > Statistics > Stat-calc
> one-variable stat

Quartiles - the % percentiles (5 number summary)

- distribution of data
- Min = minimum
 - Max = maximum
 - Q_1 = 25th percentile (median between min and median)
 - Q_2 or median = 50th percentile
 - Q_3 = 75th percentile (median between median and max)

Interquartile Range (IQR) - $IQR = Q_3 - Q_1$

Outliers - Terms with values way too high or low

* $x_i < Q_1 - 1.5 \times IQR$ or $x_i > Q_3 + 1.5 \times IQR$

Statistic Graphing techniques

Box-and-whisker plot - Describes distribution of dataset (5 # summary)



Data can be classified into two categories

1. Discrete Data - Bar charts (discrete categorical data)



2. Continuous Data - Histograms (distribution of continuous numerical data with adjacent bars)



Cumulative frequency graphs
(where vertical axis is the cumulative frequency of terms with values less than the current)

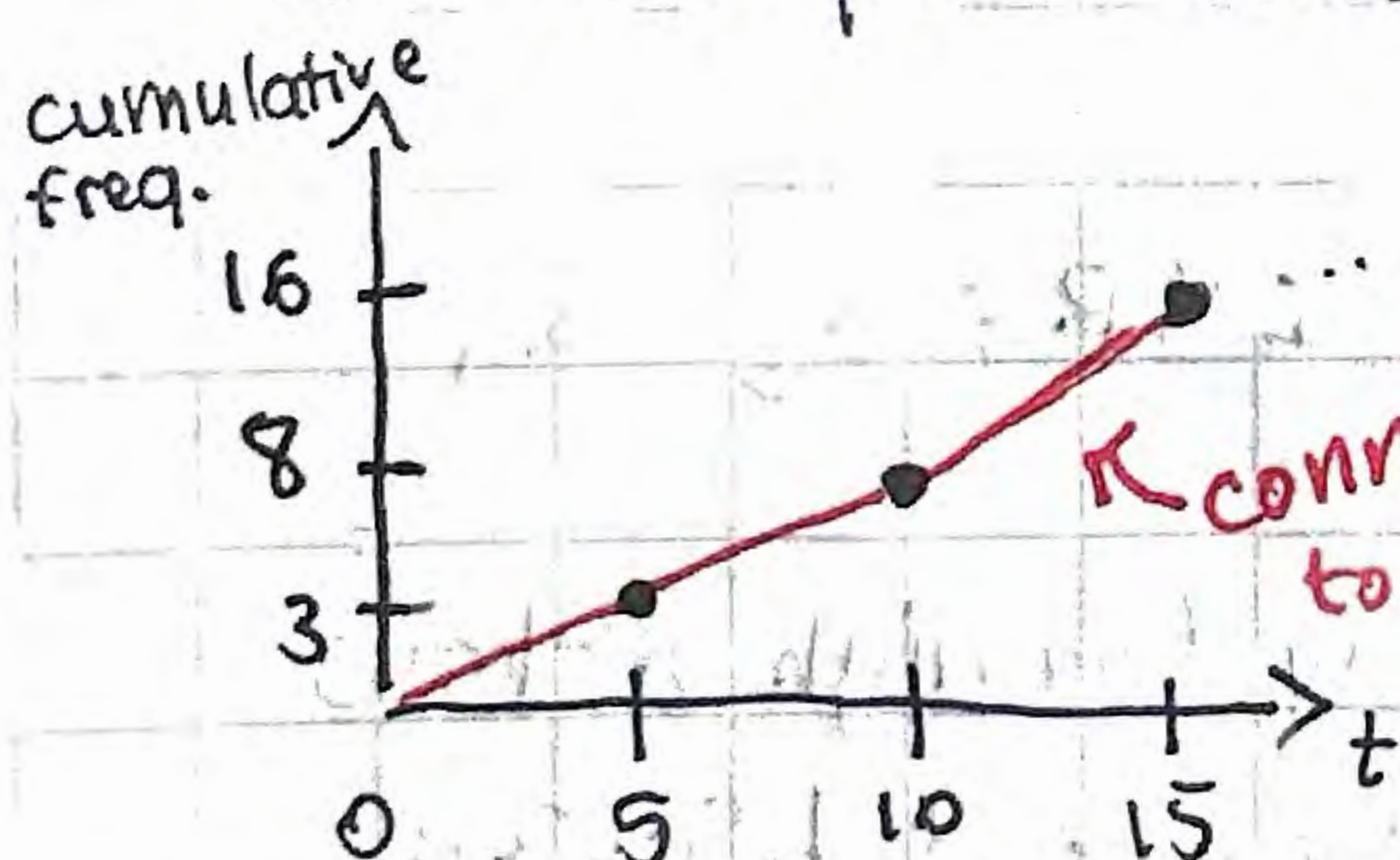
Cumulative frequency data

E.g.

Time (h)	f	Cumulative
0 to 5	3	3
5 to 10	5	8
10 to 15	8	16
...	...	...

← assume every datapoint in the given interval is the median (i.e. 3 of 2.5 h)

You can plot this in a graph



connect discrete points to form continuous data.

Calculator:

Enter one column value, & another column frequency

Menu > Stat. > Stat Calc.

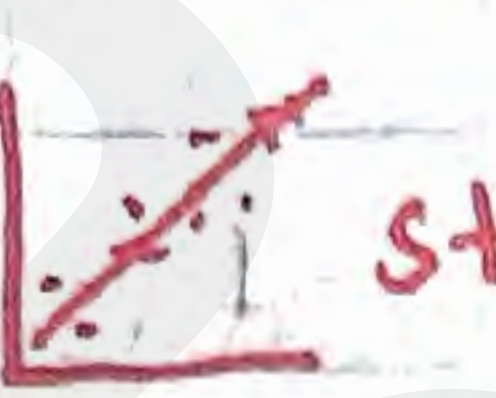
> One Var Stat > 1 list

> X1 list = values

freq. list = frequencies

Correlation refers to the relation between two variables.

The correlation coefficient, r , describes how strong the correlation is

$-1 \leq r \leq 1$ <u>where...</u>		$0 \leq r \leq 0.25$	<u>Very weak</u>
 <u>weak</u>		$0.25 < r \leq 0.5$	<u>Weak</u>
 <u>strong</u>		$0.5 < r \leq 0.75$	<u>Moderate</u>
 <u>positive ($r > 0$)</u>	 <u>negative ($r < 0$)</u>	$0.75 < r \leq 1$	<u>Strong</u>

Some relationships might not be linearly related, but that doesn't indicate a lack of correlation.

In order to calculate r , we use a graphing calculator
* Enter data (where each row must contain the two corresponding values)

label two columns
↓

" menu > Statistics > Stat Calculation > Linear Regression "

Extrapolation - When you make predictions with a value outside of the boundaries in a regression.

(i.e. if $x \in \{1, 2, 3, 4\}$ and $y \in \{2, 5, 6, 8\}$, you cannot use $x=0$ or any $x < 1$, $x > 4$ in regression)

• Regression only applies to the domain the independent is given

A regression like $y = mx + b$
↙ m means for every increase in x , an increase of m in y is expected

Regression

A regression equation is not invertible.

So a regression where x is independent, y is dependent should not be used as x is dependent, y is independent.

↳ i.e. $y = mx + b$; $x = \frac{1}{m}y - \frac{b}{m}$ will not work

Also, a linear regression equation must always pass through $(\bar{x}, \bar{y})$.

Counting

Fundamental principle - Finding total number of combinations or options given a sequence of choices involves multiplying # of options for each choice together

- ↳ Also known as multiplication rule
- ↳ Used in situations involving sequential choices

There are two main types of counting helps determine # ways to arrange / select items

- ↳ E.g. n ways for 1st choice, m ways for 2nd then total combination is $m \times n$ for both choices

1. Permutation

- Order does matter, $\{1, 2, 3\}$ is not the same as $\{2, 1, 3\}$

- Denoted by nPr

$$nPr = \frac{n!}{(n-r)!}$$

! is factorial

where... $n = \text{total \#}$
 $r = \text{\# chosen}$

easily simplified
since it's just $(n)(n-1)(n-2)\dots(n-r+1)$

- A special type of permutation is letter permutation

$$nPr = \frac{n!}{a!b!\dots}$$

where $n = \text{total \# of letters}$
 $a, b, \text{etc.} = \text{\# of each letter}$
 $a+b+\dots \text{etc.} = n$

* Can be used for non-letter problems as well

2. Combinatorics

- Order does not matter, $\{1, 2, 3\}$ is the same as $\{2, 1, 3\}$

- Denoted by nCr or $\binom{n}{r}$

$$nCr = \binom{n}{r} = \frac{n!}{(n-r)!r!} = \frac{nPr}{r!}$$

comes from dividing combinations of same element but different order

(since factorial is $r!(r-1)!\dots(1)!$)

2. Combinatorics

The formula $\binom{n}{r} = \frac{n!}{(n-r)!r!}$ also corresponds to rows of the pascal triangle

$$\begin{array}{ccccccc} & & 1 & & & & \\ & & & 1 & & & \\ & 1 & & 2 & & 1 & \\ 1 & & 3 & & 3 & & 1 \\ & 1 & & 4 & & 6 & & 4 & & 1 \end{array}$$

etc.

one way to arrange x items

Factorial - How many ways to arrange x terms is $x!$. Thus $0! = 1$

Division and subtraction in counting

- break it down case by case!
- part of casework
- Division - Dividing combinations into groups or to remove repeated elements (remove as in previous combinations as had some form of repetition, e.g. $x!$ for x_0, x_1 , etc.)
 - Multiplication - To introduce repetition or order matters or add more multiple cases
 - Subtraction - To directly remove scenarios (to meet restrictions given by question)
 - Addition - To add scenarios or more combinations

* Principle of inclusion and exclusion can be used to avoid repeats
Other than choosing elements, you may also be asked to split elements into groups

Theorem 1: n items into k groups
(where each group must have ≥ 1 element(s))

Stick and Stone counting

of possibilities = $\binom{n-1}{k-1}$
 $\downarrow$ k groups so $k-1$ dividers (think of adding dividers to split into groups)
 $\cdot \cdot | \cdot \cdot | \cdot \cdot$, $n-1$ places to add dividers (excludes ends)

Stick and stone counting

Theorem 2: n items into k groups
(where groups may have no elements)

$$\# \text{ of possibilities} = \binom{n+k-1}{k-1}$$

similar to theorem 1, but consider cases where
dividers are on the very edge

↳ $k-1$ dividers

↳ $n + k - 1$ places

e.g.

Extension

$$\binom{n+k-1-2r}{k-1}$$

is min. # of items
in i th set

Binomial Theorem

- Fast way to perform a binomial expansion
- There's a pattern we can utilize

$$(x+y)^0 =$$

$$(x+y)^1 =$$

$$(x+y)^2 =$$

$$(x+y)^3 =$$

$$(x+y)^4 =$$

$$x^2 + 2xy + y^2$$

$$x^3 + 3x^2y + 3xy^2 + y^3$$

$$x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$$

$$1 \quad 2 \quad 1$$

$$1 \quad 3 \quad 3 \quad 1$$

$$1 \quad 4 \quad 6 \quad 4 \quad 1$$

Notice its connection to the pascal's triangle
and thus to combinatorics

Why? $(x+y)(x+y)(x+y) \cdots (x+y)$

group x^n , $x^{n-1}y$, $x^{n-2}y^2$, etc.

how many ways to choose (the y)?

$$\binom{n}{0}$$

$$\binom{n}{1}$$

$$\binom{n}{2}$$

n total,
choose 0 y

n total,
choose 1 y

n total,
choose 2 y

Binomial Theorem

Generalizing this behavior, we get...

$$(x+y)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} y^r$$

$$= x^n + \binom{n}{1} x^{n-1} y + \binom{n}{2} x^{n-2} y^2 + \dots + \binom{n}{r} x^{n-r} y^r + \dots + y^n$$

In turn, we can also find the r th term

$$r^{\text{th}} \text{ term of } (x+y)^n = \binom{n}{r} x^{n-r} y^r$$

$$= \frac{n!}{(n-r)!r!} \cdot x^{n-r} y^r$$

and the constant term

(when variables cancel out)

when y is a constant, then the constant term of $(x+y)^n$ expansion is $(n+1)^{\text{th}}$ term

when x is a constant, then the constant term of $(x+y)^n$ is the 1st term ($r=0$)

but when neither is a constant, then...

you must guarantee the variables in $x^{n-r} y^r$ will cancel out

$$\text{E.g. } (3x^3 - \frac{2}{x})^8$$

$$\hookrightarrow \binom{8}{6} (3x^3)^a \left(-\frac{2}{x}\right)^b \quad \begin{array}{l} \nearrow \therefore a=2, b=6 \\ \downarrow a+b=n, a \cdot 3 = 1 \cdot b \end{array}$$

Complex Numbers

At its simplest form, complex numbers consist of a real part and an imaginary part:

$$Z = a + bi \quad Z^* = a - bi$$

↖ its conjugate
group Re and Im separately

• Real goes with real, imaginary goes with imaginary $\Rightarrow Z_1 + Z_2 = (a_1 + a_2) + (b_1 + b_2)i$

We usually may see this in quadratics and other polynomials

Conjugate root theorem

if z is a root of some polynomial, so is z^* .

This is especially useful when combined with Vieta's formulas

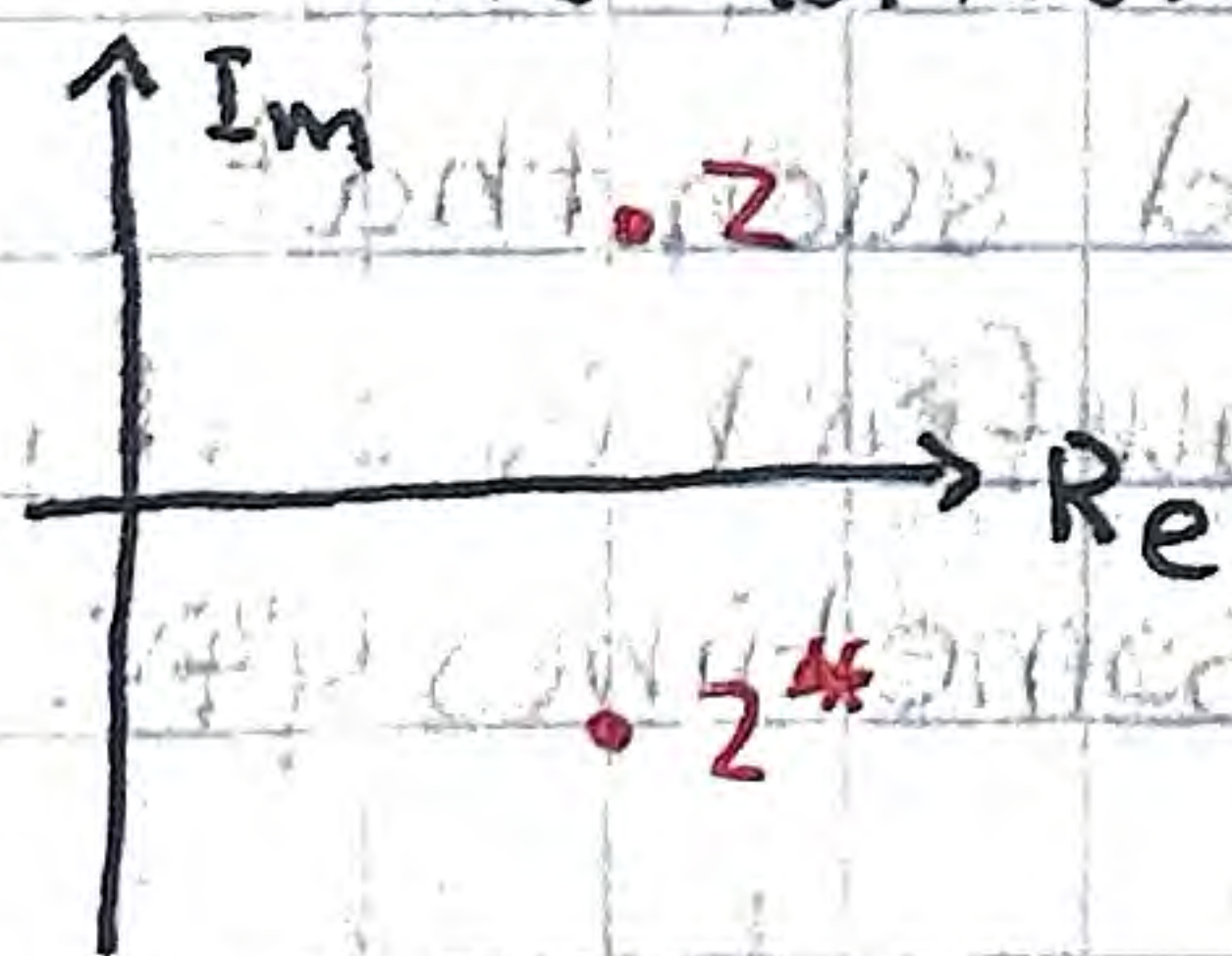
$$\text{Sum of roots} = -\frac{a_{n-1}}{a_n}$$

$$\text{Product of roots} = (-1)^n \frac{a_0}{a_n}$$

Plotting complex numbers

• The vertical axis represents the imaginary part

• The horizontal axis represents the real part



z^* is just z reflected across the Real axis

↳ Conjugates are mirrored across real axis

Product of complex numbers (rectangular)

$(a+bi)(c+di)$ may obtain i^2 , but you can simplify it

$$i = \sqrt{-1}, \quad i^2 = (\sqrt{-1})^2 = -1, \quad i^3 = i^2 \cdot i = -i, \quad i^4 = i^2 \cdot i^2 = (-1)(-1) = 1$$

↳ This means product of two complex numbers must also be simplified

Product of a complex number and its conjugate to $Z = \alpha + \beta i$

$$Z \cdot Z^* = (a+bi)(a-bi) = a^2 + b^2 = |Z|^2$$

(no i^2)

Complex numbers

Modulus

- The absolute value of the complex number, distance from the origin

Denoted as $|z|$ or $\|z\|$ or r

↳ You can use pythagorean theorem to find it

$$z = a + bi, r = |z| = \sqrt{a^2 + b^2}$$

* $r \geq 0$ always

Argument

- The angle of the complex number

Denoted as $\arg(z)$

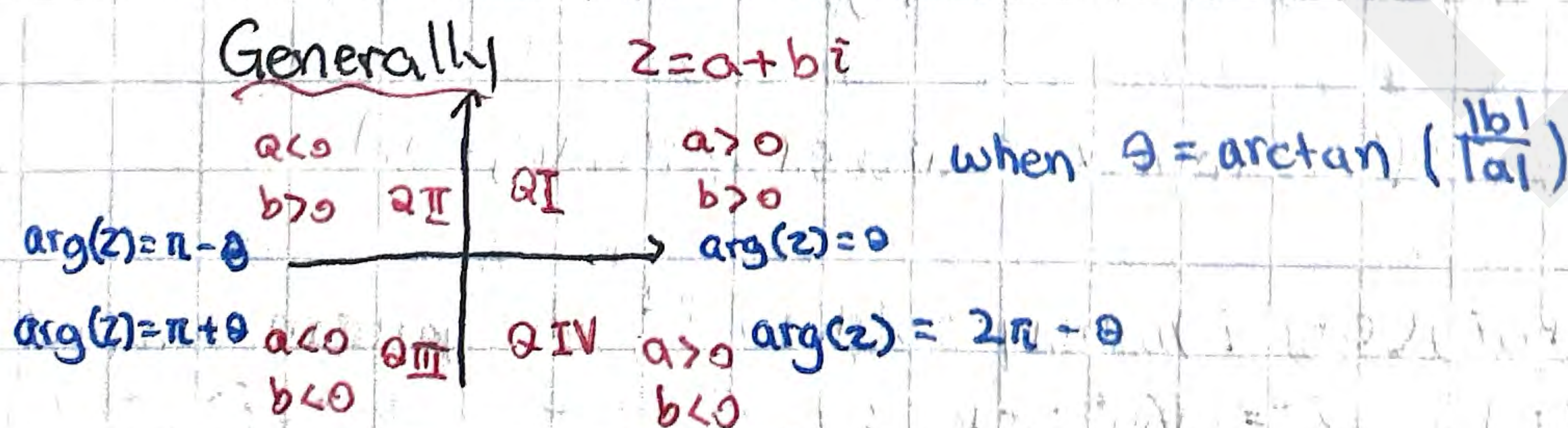
↳ You mainly use tangent to find the angle

$$z = a + bi, \arctan\left(\frac{b}{a}\right)$$

* Be aware of quadrants

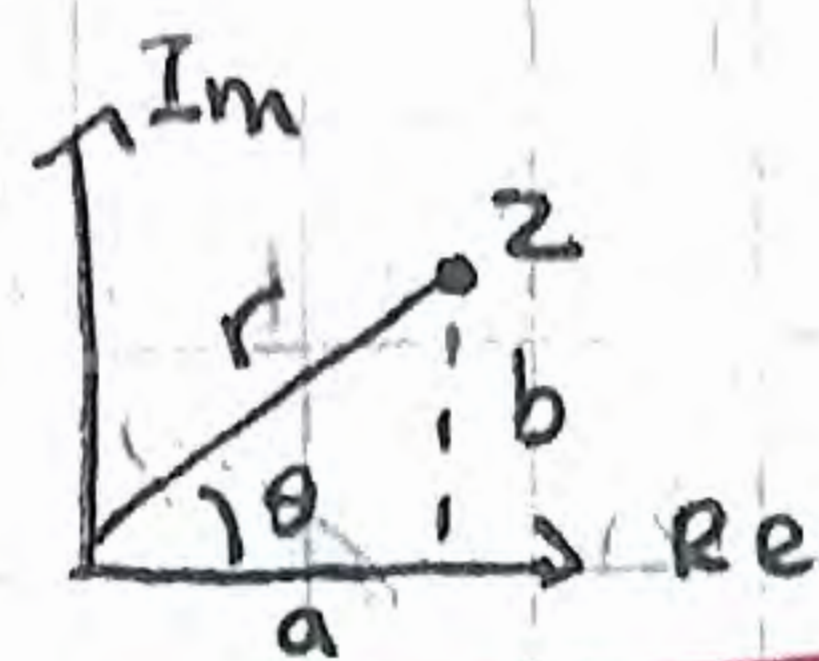
- Since $\arctan$ is only defined such that

$-\pi < \arctan(\theta) < \pi$, the angle you find is only reflective of quadrant I and sometimes QIV.



Complex numbers

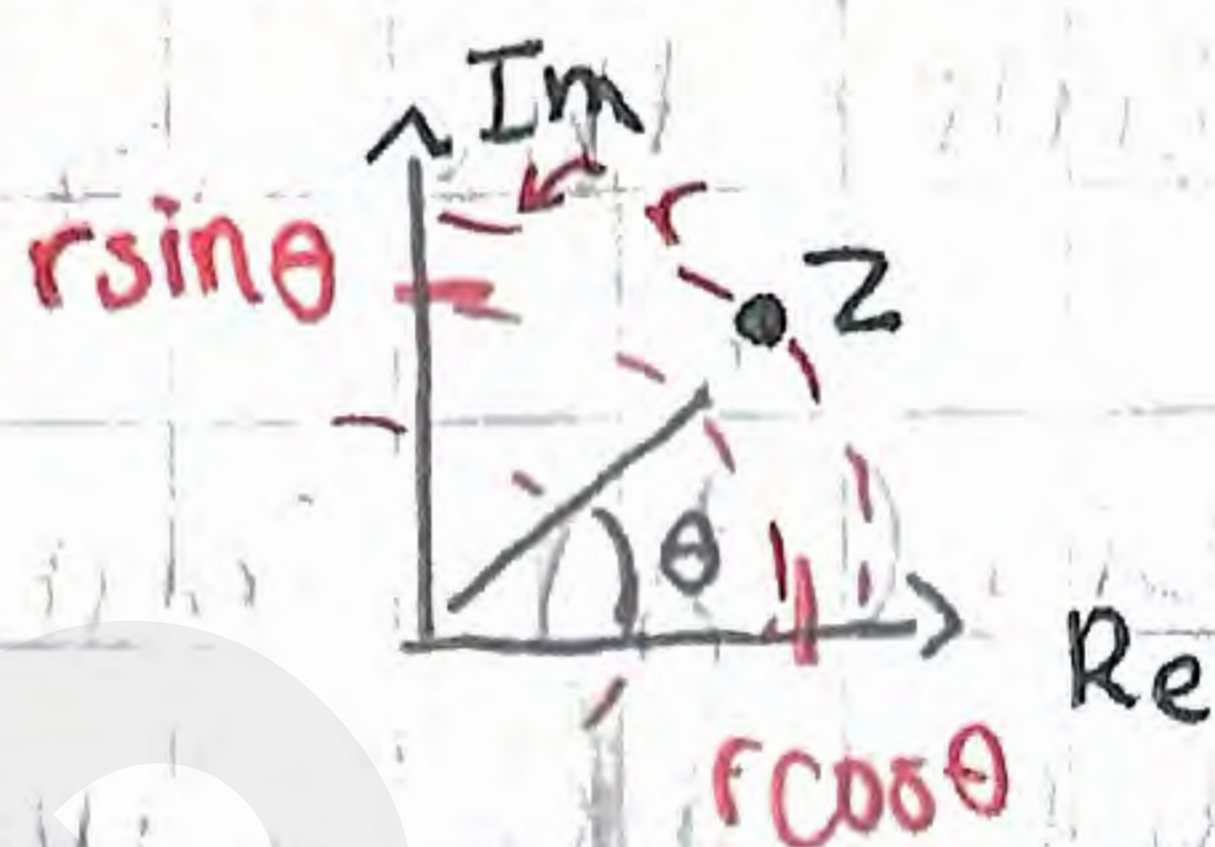
Rectangular form:



$$z = a + bi$$

we notice $\sin \theta = \frac{b}{r}$, $\cos \theta = \frac{a}{r}$

Polar form: $z = r \cos \theta + i r \sin \theta = r(\cos \theta + i \sin \theta)$



converting rectangular to polar is just using $r \sin \theta = b$, $r \cos \theta = a$

* more on graphing and conversion between forms later

Exponential form: $z = r e^{i\theta}$

we can derive it as follows:

$$\frac{dz}{d\theta} = i \cdot z$$

$$z = r(\cos \theta + i \sin \theta)$$

$$z' = r(-\sin \theta + i \cos \theta)$$

$$= r i (\cos \theta - \frac{\sin \theta}{i})$$

$$= r i (\cos \theta + i \sin \theta)$$

$$= r i \cdot z$$

$$\frac{1}{i} = \sqrt{\frac{1}{-1}} = -i$$

$$\int \frac{1}{z} dz = \int i d\theta$$

$$\ln |z| = i \theta + c$$

$$z = k e^{i\theta}$$

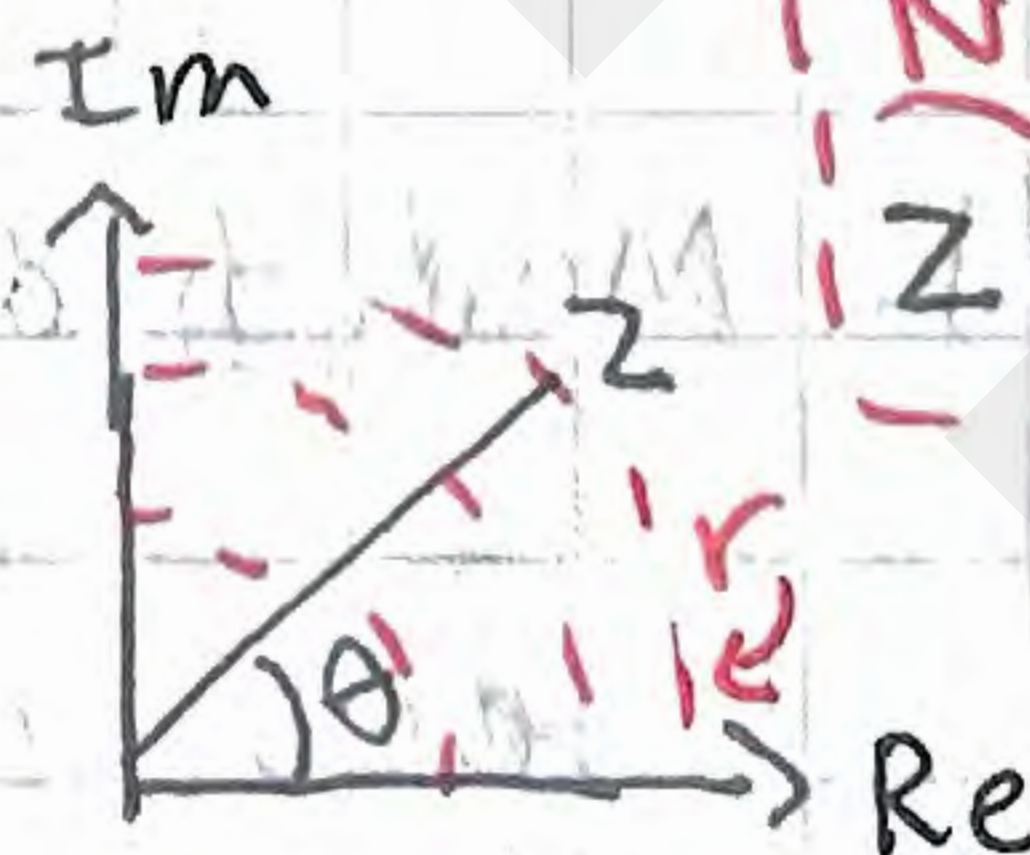
solve for k...

$$\text{e.g. } \theta = 0, z = r$$

$$z = k e^{i \cdot 0} = k$$

$$\text{thus } k = r$$

$$\Rightarrow z = r \cdot e^{i\theta}$$



NOTE:

$$z + z^* = 2r \cos(\theta)$$

Multiplying complex numbers

Doing products in rectangular is tedious
instead, use polar or exponential

* more on this later

$$z_1 z_2 = r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)) = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} (\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)) = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

sum exponents

Complex numbers

Exponents

Demoivre's theorem

$$z^n = (r \operatorname{cis} \theta)^n = r^n \operatorname{cis}(n\theta)$$

$$z^n = (re^{i\theta})^n = r^n e^{in\theta} \\ = r^n (\cos(n\theta) + i \sin(n\theta))$$

we can also take the roots

Roots of
complex #s

$$z^{\frac{1}{n}} = r^{\frac{1}{n}} \operatorname{cis}\left(\frac{\theta}{n}\right) \text{ but this isn't the only root (think polynomial)}$$

E.g. if given $z^n = r \operatorname{cis}(\theta)$
then there must be n roots/solutions
to z

To find all n solutions, there are two ways
All ways require $r^{\frac{1}{n}}$ as new modulus
but angles change

1. Manipulating the argument

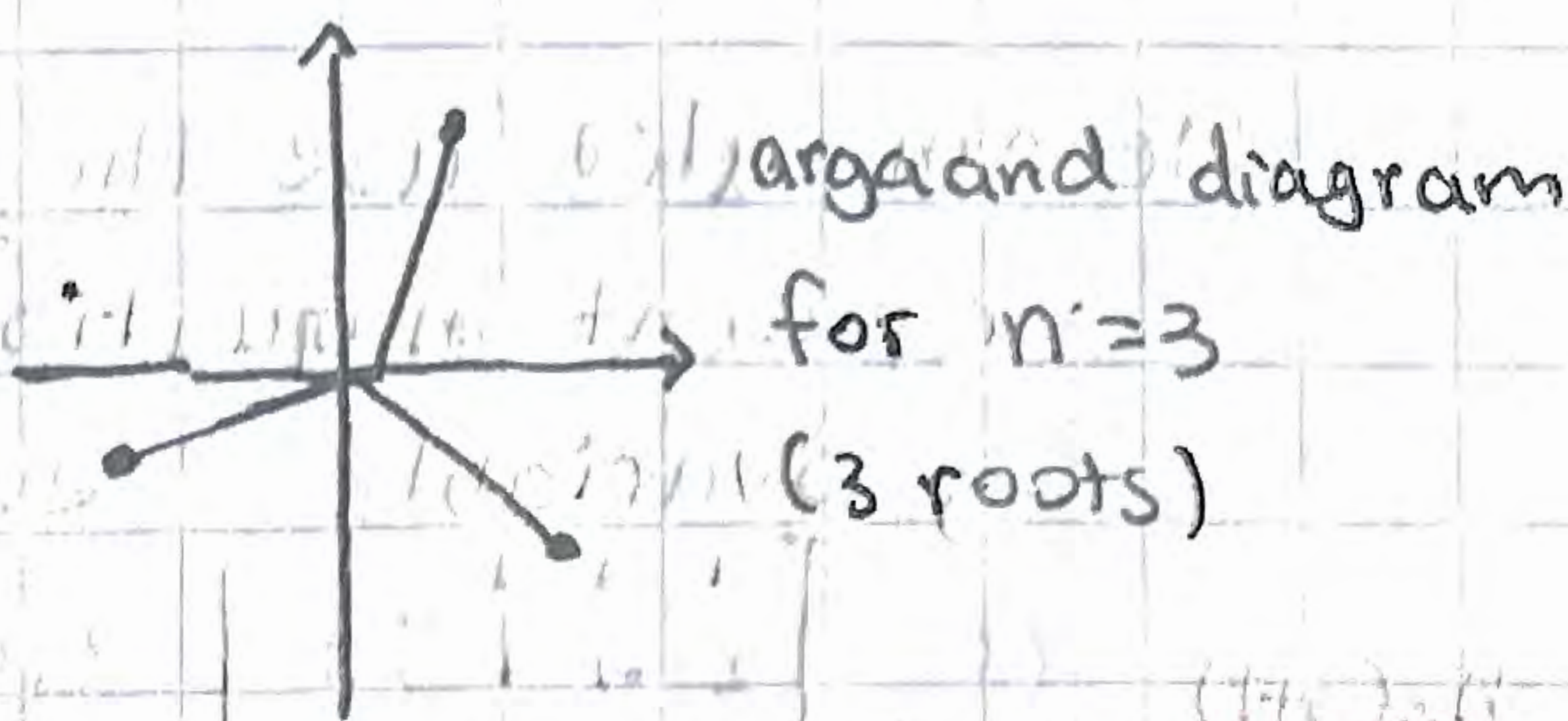
Since $\operatorname{trig}(\theta) = \operatorname{trig}(\theta + 2\pi k)$ for $k \in \mathbb{Z}$
that means that we should check $\frac{\theta + 2\pi k}{n}$
as modulus for each root (it's valid if 0 to 2π)

E.g. $z^3 = 8 \operatorname{cis} \frac{\pi}{3}$ but could also be $8 \operatorname{cis} \left(\frac{\pi}{3} + 2\pi \right)$ or $8 \operatorname{cis} \left(\frac{\pi}{3} + 4\pi \right)$
so $z = 2 \operatorname{cis} \left(\frac{\pi}{9} \right)$ or $2 \operatorname{cis} \left(\frac{7\pi}{9} \right)$ or $2 \operatorname{cis} \left(\frac{13\pi}{9} \right)$

Complex numbers

2. Argand diagram

- The angle between all n roots are equal and form a circle



- Thus you can find first/simplest root and then add $\Delta\theta$ where $\Delta\theta = \frac{2\pi}{n}$

E.g. $z^3 = 8 \text{cis } \frac{\pi}{3} \Rightarrow z = 2 \text{cis} \left(\frac{\pi}{3} \right) + \Delta\theta$
 $\Delta\theta = \frac{2\pi}{3}$ thus z also $= 2 \text{cis} \left(\frac{7\pi}{3} \right)$ or $2 \text{cis} \left(\frac{13\pi}{3} \right)$

Solving n variable equations (linear) with matrices

You can represent n variable linear equations as a matrix which can facilitate solving equations

E.g. 3 variable, 3 equation:

$n \times n$
matrix

$$x + y + z = 1$$

$$2x - y + z = 5$$

$$-3x + 2y - 4z = 7$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ -3 & 2 & -4 \end{bmatrix}$$

each row represents one equation
each column represents one variable

So multiplying the coefficient matrix by a variable matrix should equal the constant matrix

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ -3 & 2 & -4 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}$$

when $A \cdot X = B$, $X = B \cdot A^{-1}$

note that there's no division with matrices

A^{-1} refers to "inverse"

so plugging $\begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ -3 & 2 & -4 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}$ gives $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ in calculator

n variable (linear) equations with matrices

We can also use the Reduced Row-Echelon Form

• represent n equations as one single matrix

(E.g. 3 variable 3 equation

$n \times (n+1)$
matrix

$$\begin{bmatrix} 1 & 1 & 1 & 5 \\ 2 & -1 & 1 & 5 \\ -3 & 2 & 4 & 7 \end{bmatrix}$$

coefficients and constant term

$$\begin{matrix} \uparrow & \uparrow & \uparrow & \uparrow \\ x & y & z & a \end{matrix}$$

you can multiply rows by constants, etc. until you end up with the wanted form

This method involves adding rows to each other to force the matrix into the form

basically same way as elimination $\rightarrow$ in algebra, but with matrices

$$\begin{bmatrix} 1 & 0 & 0 & x \\ 0 & 1 & 0 & y \\ 0 & 0 & 1 & z \end{bmatrix}$$

where x, y, z would be the solutions

in a calculator it's

$$\text{rref}\left(\begin{bmatrix} 1 & 1 & 1 & 5 \\ 2 & -1 & 1 & 5 \\ -3 & 2 & 4 & 7 \end{bmatrix}\right)$$

Root of unity

nth root of unity is $z^n = 1$ for $n \in \mathbb{Z}^+$

• the solutions are evenly spread upon a unit circle

• so $z_k = e^{\frac{2\pi i k}{n}}$ for $k \in \{0, 1, 2, \dots, n-1\}$
 z_0 is always 1

all roots of unity have $|z| = 1$

*some of the roots are conjugates

$$e^{\frac{2\pi i k}{n}} \text{ and } e^{\frac{2\pi i (n-k)}{n}}$$

This is useful, especially in factorizations

roots of unity

$$z^n - 1 = (z-1)(z-z_1)(z-z_2)\dots$$

you can multiply the two roots as $z-z_k$ to get

$$\Rightarrow z^2 - 2z \cos\left(\frac{2\pi k}{n}\right) + 1$$

in this case
 $z+z^* = 2\cos$
 $z \cdot z^* = |z|^2 = 1$

Complex numbers and their conjugates

Recalling
and

$$\begin{aligned} z + z^* &= 2 \cdot r \cdot \cos \theta \\ z \cdot z^* &= r^2 \end{aligned}$$

We can consider... $(x+z)(x-z^*)$
which must then be

$$x^2 - 2 \cdot r \cdot \cos(\theta)(x) + r^2$$

Complex numbers and basic operations

group Im

Addition:
Subtraction

$$(a+bz) + (c+dz) = (a+c) + (b+d)z$$

group Re

$e^{ix} + e^{i\beta}$ are difficult to add

$\cos x + \cos y$
is hard to
combine
into one single
function

→ Polar and exponential are difficult to add and divide

However, there are some special cases:

$$\begin{aligned} z + z^* &= (a+bz) + (a-bz) = \underline{2a} \quad \text{rectangular} \\ &= r(\cos \theta + i \sin \theta) + r(\cos(2\pi - \theta) + i \sin(2\pi - \theta)) \\ &= \underline{2r \cos \theta} \quad \text{polar} \\ &= re^{i\theta} + re^{-i\theta} = \underline{r \cdot (e^{i\theta} + e^{-i\theta})} \quad \text{exponential} \end{aligned}$$

Multiply: $(a+bz)(c+dz) = ac + adz + bcz - bd = (ac-bd) + (ad+bc)z$
rectangular

$$\begin{aligned} &r_a(\cos \alpha + i \sin \alpha) \cdot r_b(\cos \beta + i \sin \beta) \\ &= r_a \cdot r_b \cdot (\cos \alpha \cos \beta - \sin \alpha \sin \beta + i \sin \alpha \cos \beta + i \sin \beta \cos \alpha) \\ &= r_a \cdot r_b (\cos(\alpha + \beta) + i \sin(\alpha + \beta)) \quad \text{polar} \end{aligned}$$

$$r_a e^{i\alpha} \cdot r_b e^{i\beta} = r_a \cdot r_b \cdot e^{i(\alpha + \beta)} \quad \text{exponential}$$

Complex numbers and basic operations

Division: $\frac{a+bz}{c+dz}$ the trick is to use conjugates...

$$= \frac{a+bz}{c+dz} \cdot \frac{(c-dz)}{(c-dz)} = \frac{(a+bz)(c-dz)}{c^2+d^2}$$

$$= \frac{(ac+bd) + (bc-ad)z}{c^2+d^2}$$

$$= \frac{r_a \cdot \text{cis } \alpha}{r_b \cdot \text{cis } \beta} = \frac{r_a}{r_b} \text{cis } (\alpha - \beta)$$

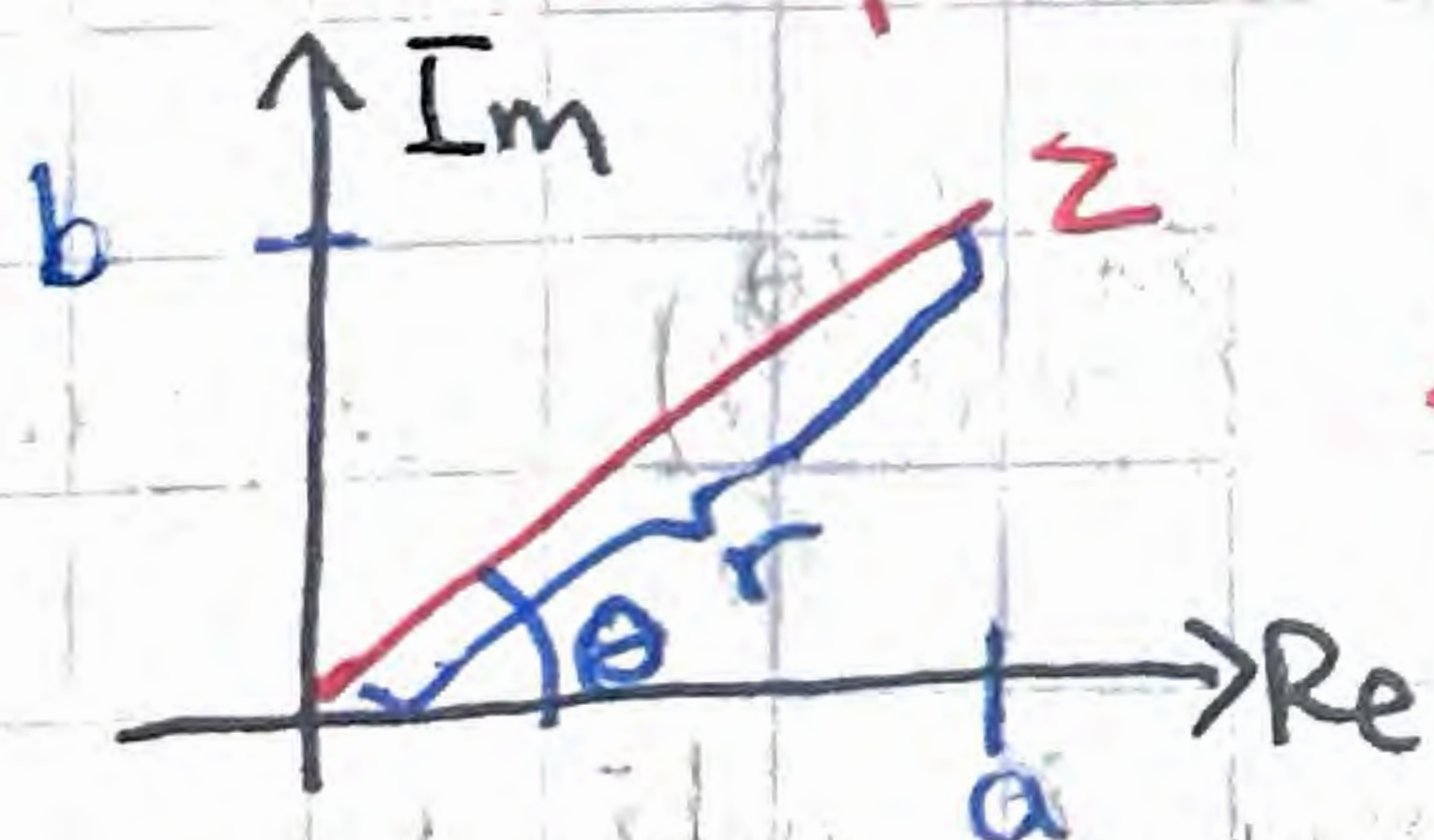
$$= \frac{r_a}{r_b} \cdot \frac{e^{i\alpha}}{e^{i\beta}} = \frac{r_a}{r_b} \cdot e^{i(\alpha-\beta)}$$

follows from exponential

Conversion between forms

$$z = a+bz = r(\cos\theta + z \sin\theta) = r e^{i\theta}$$

each form is composed of a, b or r, θ



* r is the modulus so $\sqrt{a^2+b^2}$
"hypotenuse"

θ is the argument, using trig we get...
angle of the right Δ

$$* \arctan\left(\frac{b}{a}\right) = \arcsin\left(\frac{b}{r}\right) = \arccos\left(\frac{a}{r}\right)$$

a and b are the real and imaginary parts respectively...

$\uparrow$
cos

$\uparrow$
sin

$$* a = r \cdot \cos\theta = r \cdot \frac{e^{i\theta} + e^{-i\theta}}{2}$$

recall conjugates

$\uparrow$
"legs"

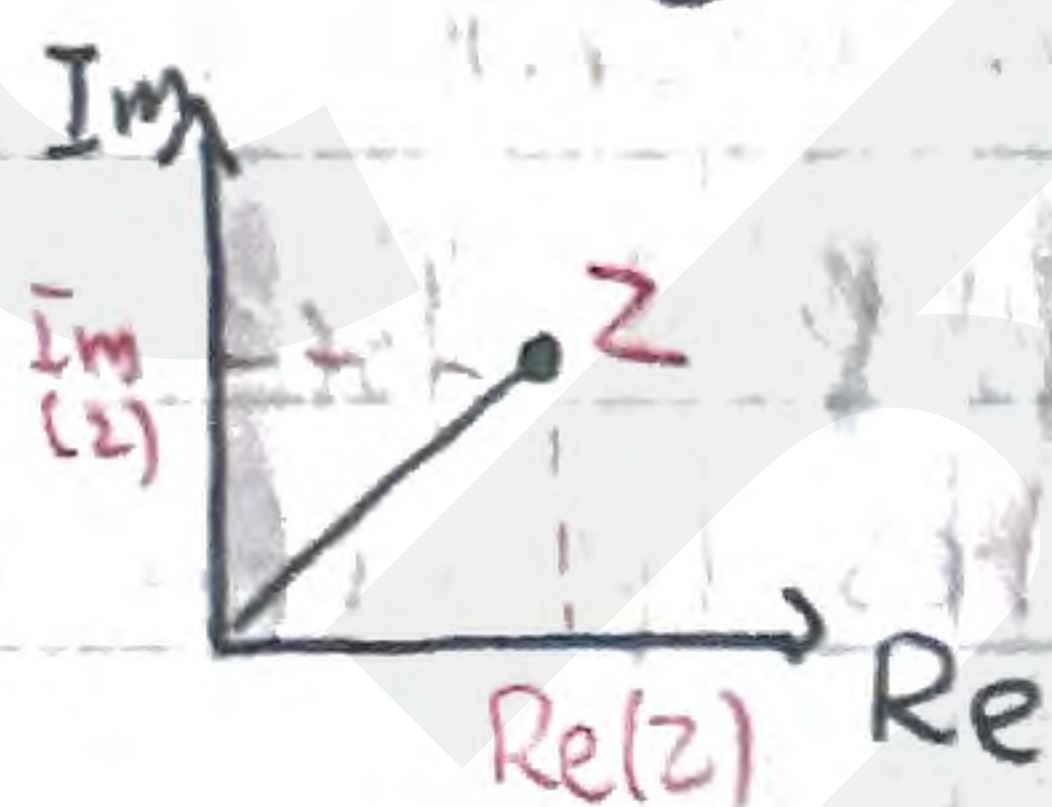
$$* b = r \cdot \sin\theta = r \cdot \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

recall conjugates

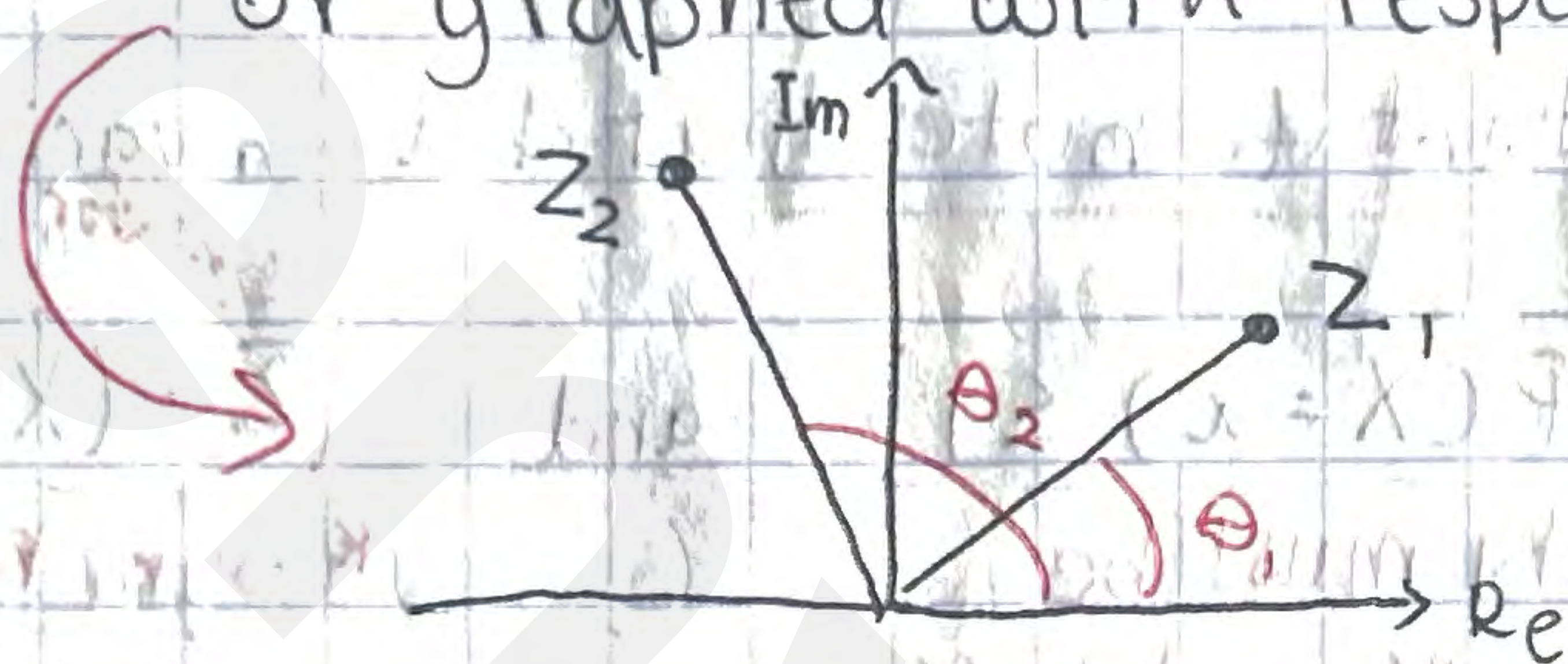
Graphing complex numbers

Rectangular form is typically easiest to graph

- Real part represents horizontal component
- Imaginary part represents vertical component



In other forms, complex numbers should either be converted to rectangular or graphed with respect to the angle



*make sure the angle starts on the Real axis and goes counterclockwise

Random Variables

Random variables are typically denoted by a capital letter

↳ A random variable X is a quantity whose value depends on a random event

↳ it's not a completely random number itself
↳ it's a model of outcomes of a random event

Notation of probability

$P(X=x)$ = probability of random variable taking on the value x ← lower case denotes specific value

Discrete Random Variable - A random variable of discrete data

$$0 \leq P(X=x) \leq 1$$

* Probability must be between 0% and 100%

and

$$\sum P(X=x) = 1$$

* Sum of probabilities must complete 100%

E.g. Probability distribution of T where T is the total score of two fair six sided die roll: one face '1', two faces '2', three faces '3'

since they are discrete a table can be used to present them

t	2	3	4	5	6
$P(T=t)$	$\frac{1}{36}$	$\frac{4}{36}$	$\frac{10}{36}$	$\frac{12}{36}$	$\frac{9}{36}$
	↑ 1+1	↑ 2+1	↑ 2+2 3+1	↑ 2+3	↑ 3+3

Thus the probability the total score is more than 4 is $\frac{12}{36} + \frac{9}{36} = \frac{21}{36}$
↳ rolling die several times

* Finding the total score probability = $\sum P_1 \cdot P_2$ etc. probability each time meets conditions

Expected value - The weighted mean of the random variable

where ...

$$E(x) = \sum x \cdot P(X=x) = \frac{\sum x \cdot \text{# of ways of } x}{\text{total # of combinations}}$$

↑
weighting

Probability

E.g. with a dice, you would be multiplying each number on side by the probability that side is rolled

↳ zero or zero sum

When an outcome is fair, its expected value is at equilibrium

Variance - Recalling from statistics, a measure of the dispersion of a set of data around its average

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= \sum x^2 \cdot P(X=x) - \left[\sum x \cdot P(X=x) \right]^2$$

weighting
squared,
probability same

square
expected
value

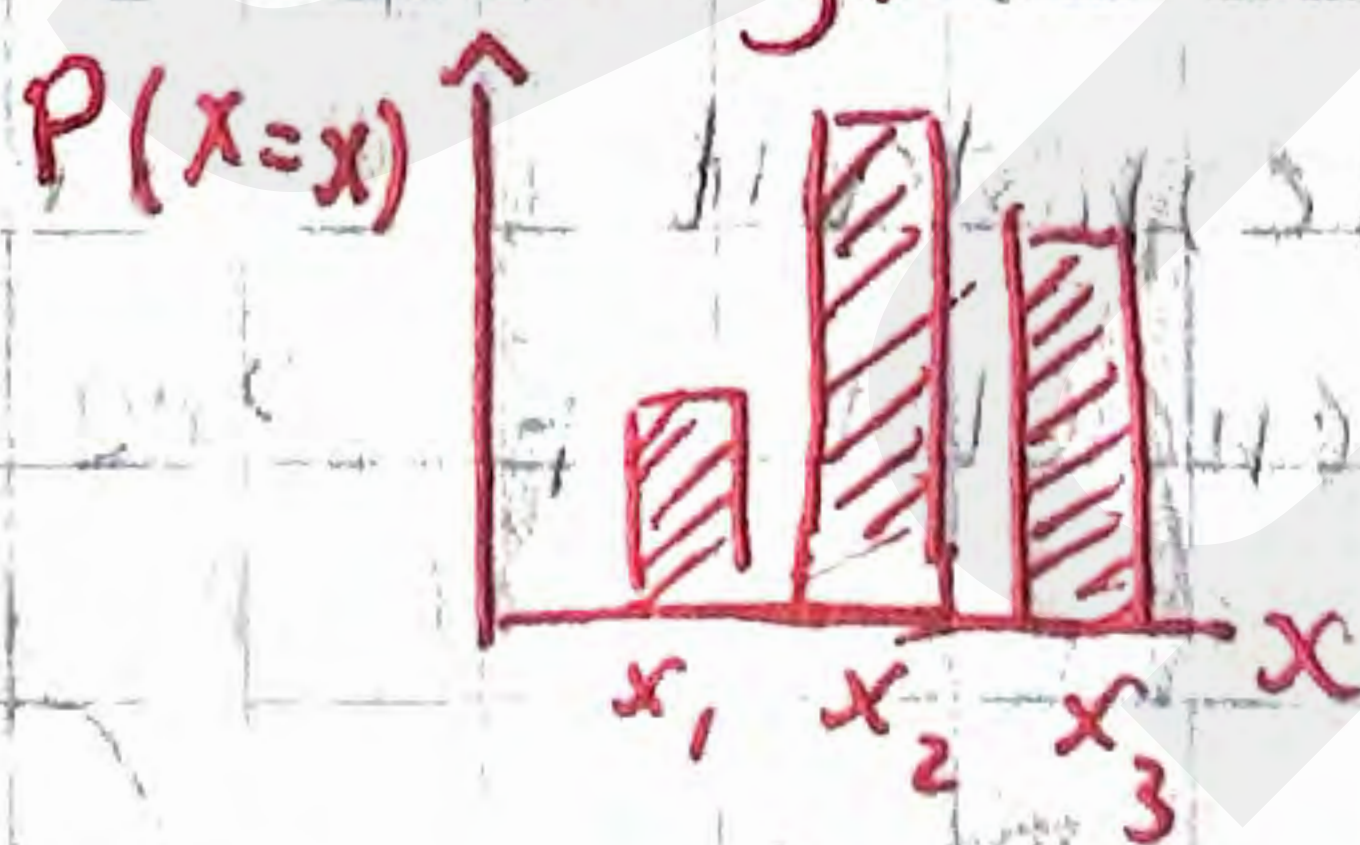
Discrete random variable can be presented with tables, graphs or probability formulas

lists possible values
and corresponding
probabilities

discrete graphs
like probability
histograms

functions
defined for
specific values
(typically integer)

x	...
$P(X=x)$	...



Binomial Distribution

Binomial - two outcomes

Binomial distribution models the results of a discrete random variable with two outcomes (e.g. success or failure)

Recall the binomial theorem...

$$(p + q)^n = \sum \binom{n}{r} p^r q^{n-r}$$

where p or $(1-p)$

in a binomial distribution

p and q denotes success and failure probability respectively (must sum to 100%)

how many ways can there be success?

$$\binom{n}{r} p^r q^{n-r}$$

denotes the probability

for exactly r successes for $p + q = 1$

probability for exactly r successes in those ways

p is probability of success
 q is probability of failure

A binomial distribution must have...

- A fixed # of trials
- Each trial is either 'success' or 'failure'
- Probability success is constant
- Trials are independent of one another

$$* X \sim B(n, p)$$

of trials $\uparrow$ probability of success $\uparrow$

"the random variable X follows the binomial distribution of n trials with a p probability of success"

Binomial distribution

Consequently, the # of successes when set as an exact value is the exponent that the probability of success must be raised to

$$\binom{n}{r} p^r q^{n-r}$$

$\binom{n}{r}$ $\rightarrow$ n choose r ways for this to occur
 p^r $\rightarrow$ probability for exactly r successes
 q^{n-r} $\rightarrow$ probability for exactly $n-r$ failures

And thus...

for # of successes greater or equal to...

$$P(X \geq r) = \sum_{i=r}^n \binom{n}{i} p^i q^{n-i}$$

or for # of successes less than or equal to...

$$P(X \leq r) = \sum_{i=0}^r \binom{n}{i} p^i q^{n-i}$$

To facilitate this process, you can use a graphing calculator

menu: $\rightarrow$ Probability $\rightarrow$ Distributions

$\rightarrow$ Binomial Pdf (Probability density function)

$\rightarrow$ Probability of exactly r successes

"binomPdf(n, p)" or "binomPdf(n, p, r)"

$\rightarrow$ returns a list of all probabilities

$\rightarrow$ returns just the probability of r successes

$\rightarrow$ Binomial Cdf (Cumulative distribution function)

$\rightarrow$ Cumulative probability from a to b # of successes

"binomCdf(n, p, a, b)"

$\rightarrow$ lower bound

Binomial Distribution

Expected value, $E(x) = n \cdot p$

↳ expected # of successes

= trials $\times$ success probability

Variance, $\text{Var}(X) = n \cdot p \cdot q$

↳ a single trial has variance $p \cdot q$

so n trials is $n \cdot p \cdot q$

Algebraic equations with binomial

Recall probability of r successes is

$$\binom{n}{r} p^r q^{n-r}$$

That means probability of not r successes is

$$P(X \neq r) = 1 - \binom{n}{r} p^r q^{n-r}$$

useful when asked $P(X \geq 1)$ so $r \neq 0$

> You can also create algebraic equalities or inequalities
E.g. Large box of flowers, $\frac{1}{4}$ red, rest are white.

How many flowers should be randomly picked so that probability of at least one red is greater than 0.95?

$$P(R \geq 1) > 0.95$$

$$1 - \left(\frac{3}{4}\right)^n > 0.95$$

↳ You can use log to solve
or table

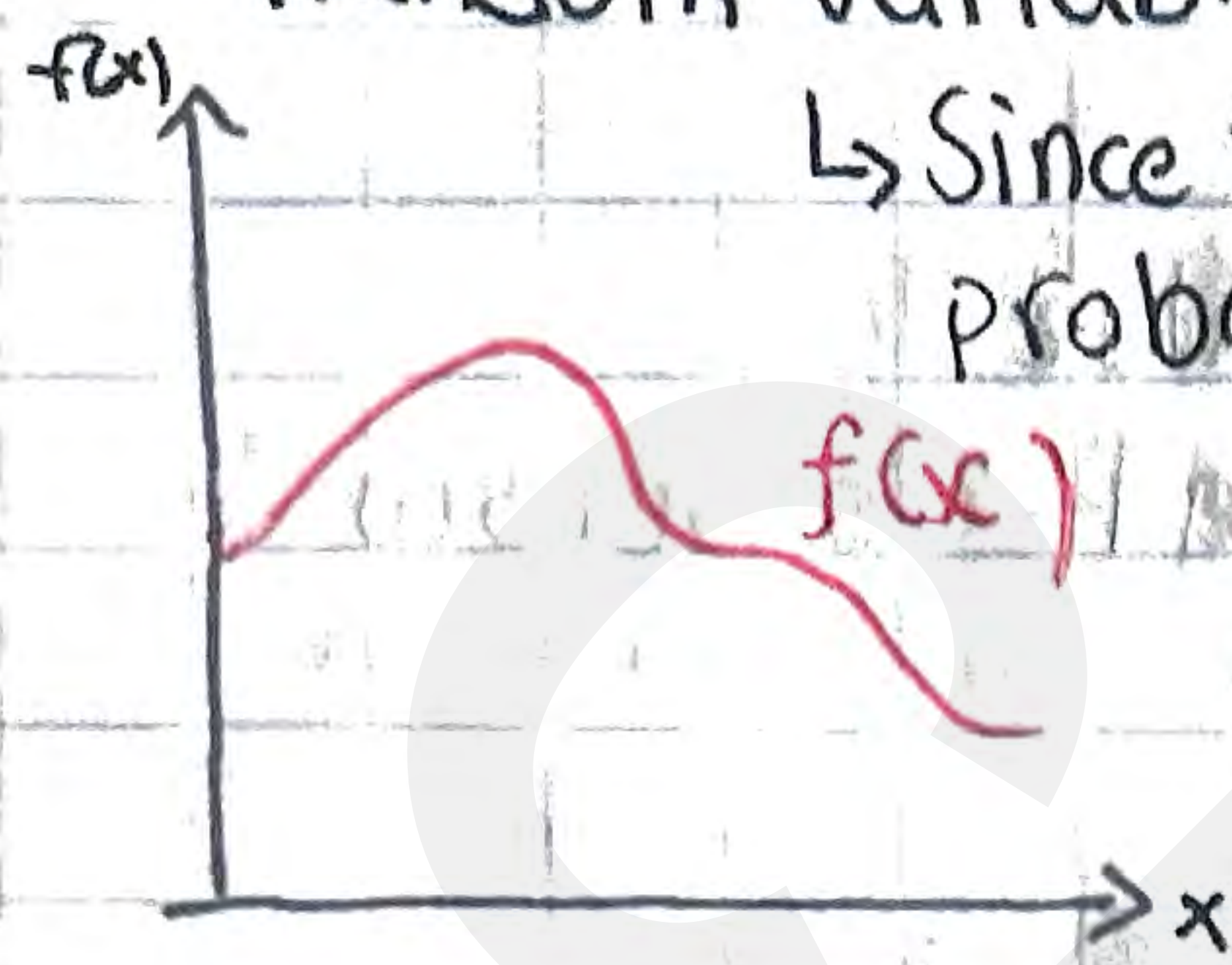
(Graph > Table > Split-screen Table)

$$\ggg n = 11$$

Continuous Random Variable

Random variables that involve a continuous dataset

↳ Since there are infinite values the variable can take on, the probability at any single, specific value is always 0

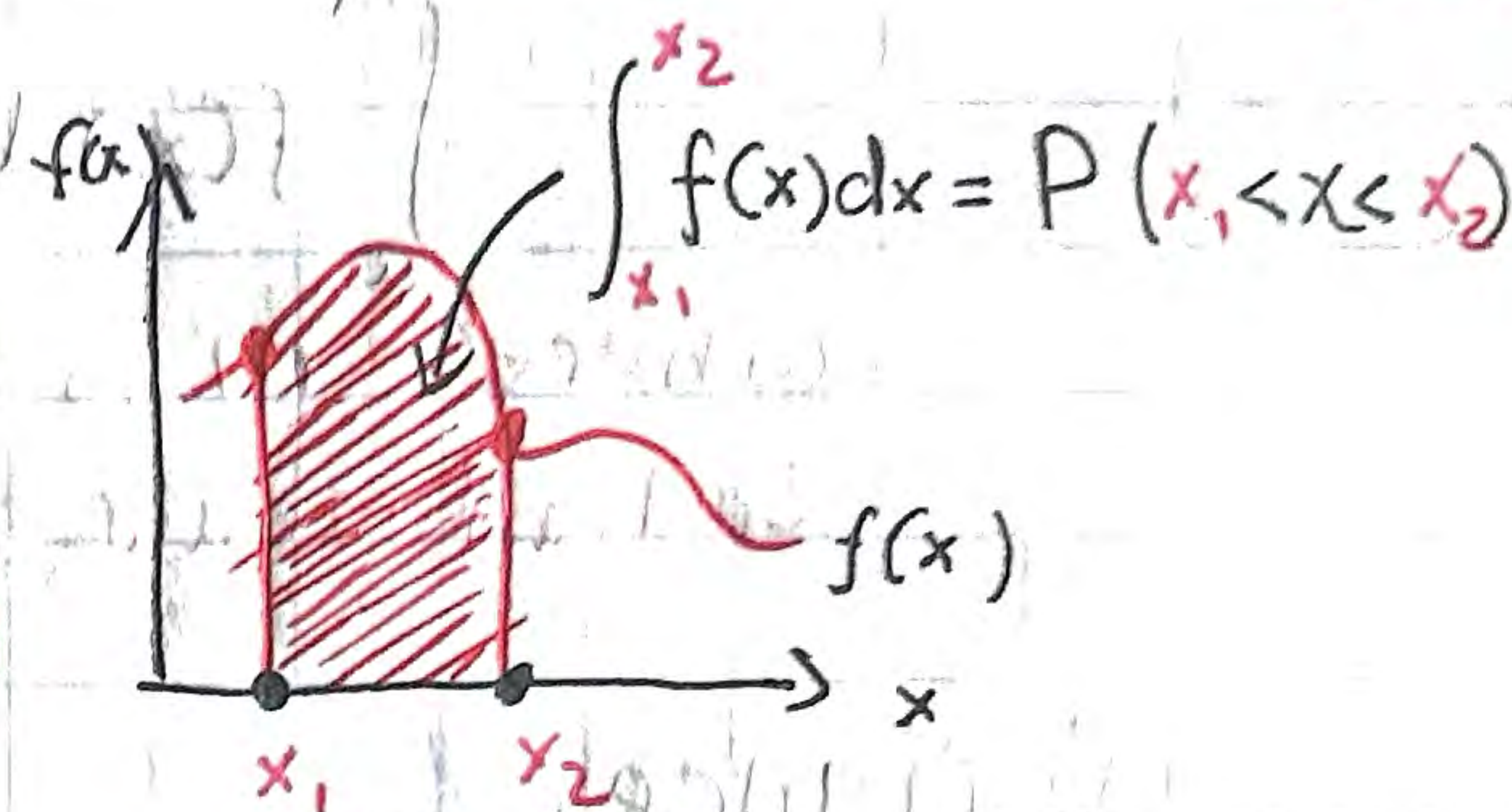


where the probability is the area under the curve

↓
 $P(X=x)=0$
for all x

Since probability must sum to 100%

① $\int_a^b f(x) dx = 1$ ← Probability of everything



Probabilities must always be positive...
thus the second condition must be

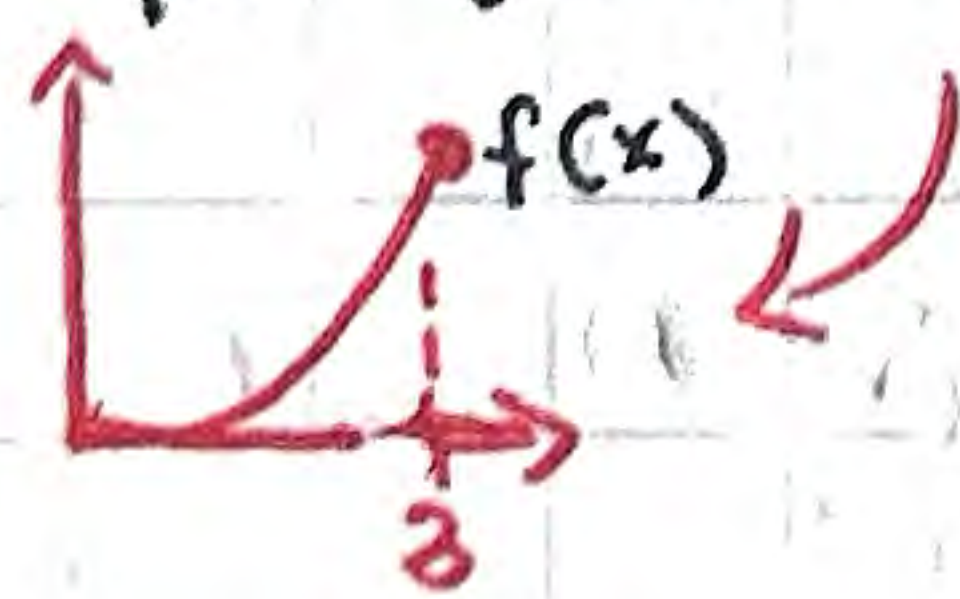
② $f(x) \geq 0$ for $a < x < b$

← a, b range at which $f(x)$ is defined
(if all $\mathbb{R}$ then $a = -\infty$, $b = \infty$)

The Mode refers to the most occurring value, this is the value when $f(x)$ is at its maximum

E.g. for $f(x) = \frac{x^2}{9}$ for $0 \leq x \leq 3$, the mode occurs at $x=3$

absolute peak means highest occurrence



The mean refers to the average, expected value.

Recalling for a discrete random variable: $\sum x \cdot P(x)$

↳ For continuous we use...

$$\int_a^b x \cdot f(x) dx$$

Continuous Random Variable

The median is the middlemost value

↳ Half the dataset is above, half is below

↳ Thus...

$$\int_a^m f(x) dx = \int_m^b f(x) dx = 0.5$$

where m is the value of the median in the dataset x

Variance

Recall $\text{Var}(x) = E(x^2) - E(x)^2$

For continuous, it is also

$$\text{Var}(x) = \int_a^b (x - \mu)^2 \cdot f(x) dx$$

↑ expected value (mean)

$$= \int_a^b x^2 \cdot f(x) dx - \mu^2$$

Transformations in random variable

	$E(x)$ expected value	$\text{Var}(x)$ variance
X	$E(X)$	$\text{Var}(x)$
aX	$a \cdot E(X)$ (mean)	$a^2 \cdot \text{Var}(x)$ (horiz. stretch, scaling by a)
$X+b$	$E(X)+b$ (shifting all up)	$\text{Var}(x)$ (equal spread, different center)
$aX+b$	$a \cdot E(X)+b$	$a^2 \cdot \text{Var}(x)$

Continuous Random variable

E.g. Let $f(x) = \frac{x^2}{9}$ defined from $0 \leq x \leq 3$

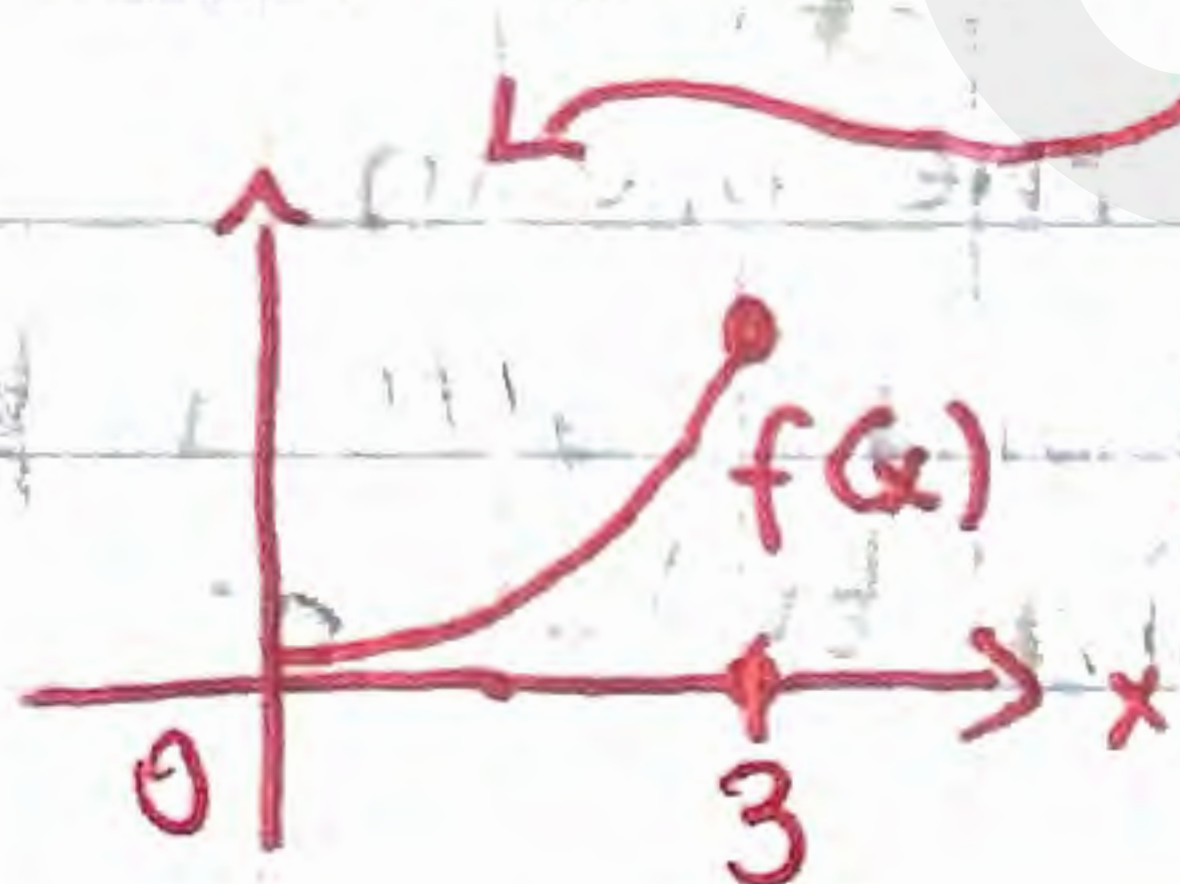
a) Can $f(x)$ model a Pdf?

① is $f(x) \geq 0$? $\checkmark$, $f(x)$ is quadratic

② is $\int_a^b f(x) dx = 1$? $\checkmark$

$$\int_0^3 f(x) dx = \left. \frac{x^3}{27} \right|_0^3 = \frac{27}{27} - 0 = 1$$

b) Find the mode, mean, and median



$$\begin{aligned} \int_0^3 x f(x) dx &= \int_0^3 \frac{x^3}{9} dx = \left. \frac{x^4}{36} \right|_0^3 \\ &= \frac{81}{36} = \frac{9}{4} \end{aligned}$$

$x = \frac{9}{4}$

$$\begin{aligned} \int_0^m f(x) dx &= \int_m^3 f(x) dx = \frac{1}{2} \\ \frac{1}{2} &= \left. \frac{x^3}{27} \right|_0^m = \frac{m^3}{27} \end{aligned}$$

$$m^3 = \frac{27}{2}, m = \sqrt[3]{\frac{27}{2}}$$

c) Find the variance

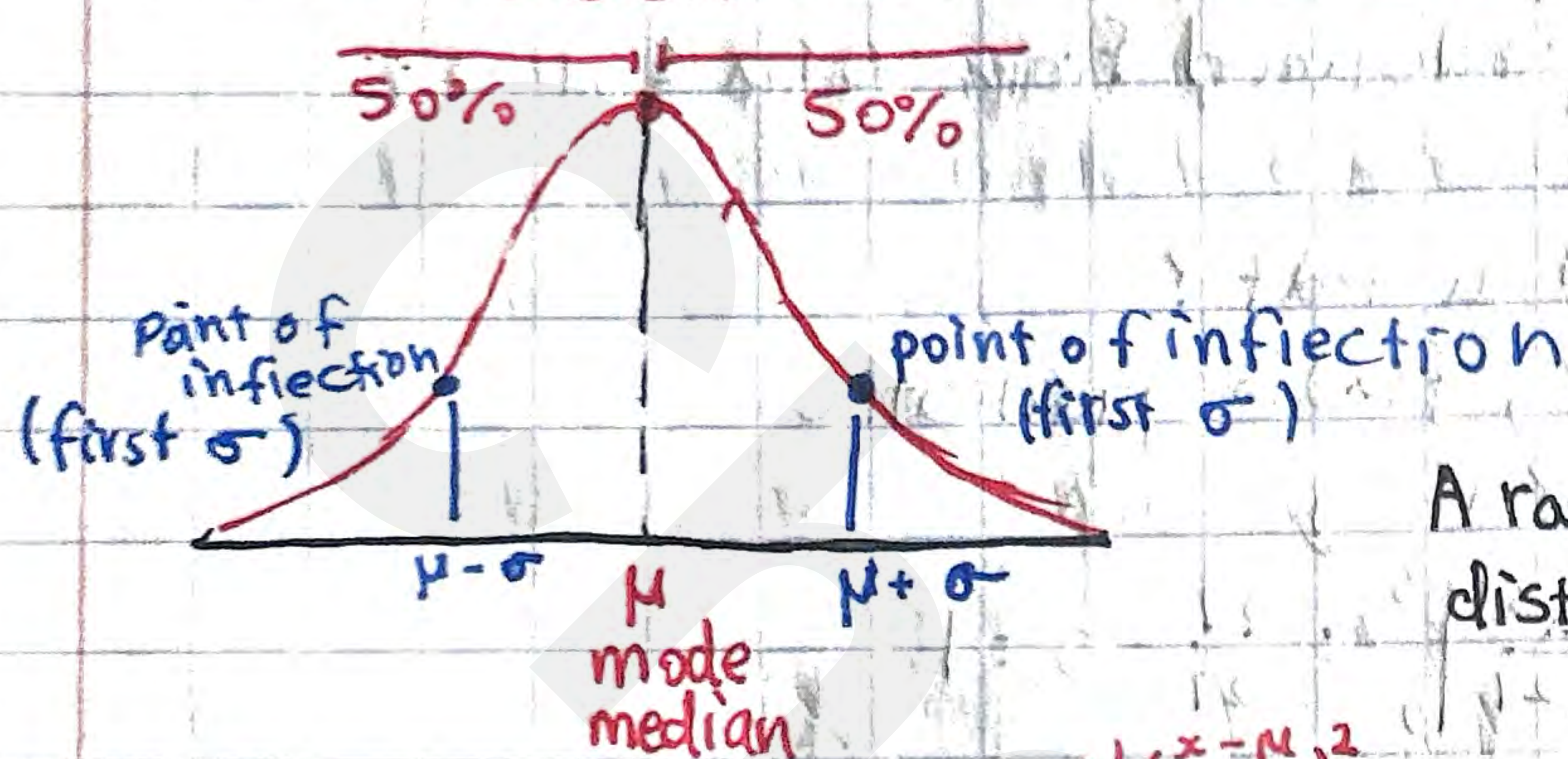
$$\begin{aligned} \text{Var}(x) &= \int_0^3 x^2 f(x) dx - E(x)^2 = \int_0^3 \frac{x^4}{9} dx - \left(\frac{9}{4}\right)^2 \\ &= \left. \frac{x^5}{45} \right|_0^3 - \frac{81}{16} \approx 0.338 \end{aligned}$$

d) Probability of the value being $1 \leq x \leq 2$?

$$P(1 \leq x \leq 2) = \int_1^2 f(x) dx = \int_1^2 \frac{x^2}{9} dx = \left. \frac{x^3}{27} \right|_1^2 = \frac{7}{27}$$

Normal Distribution

A unique type of a distribution curve where the mean, mode, and median are found at the same point



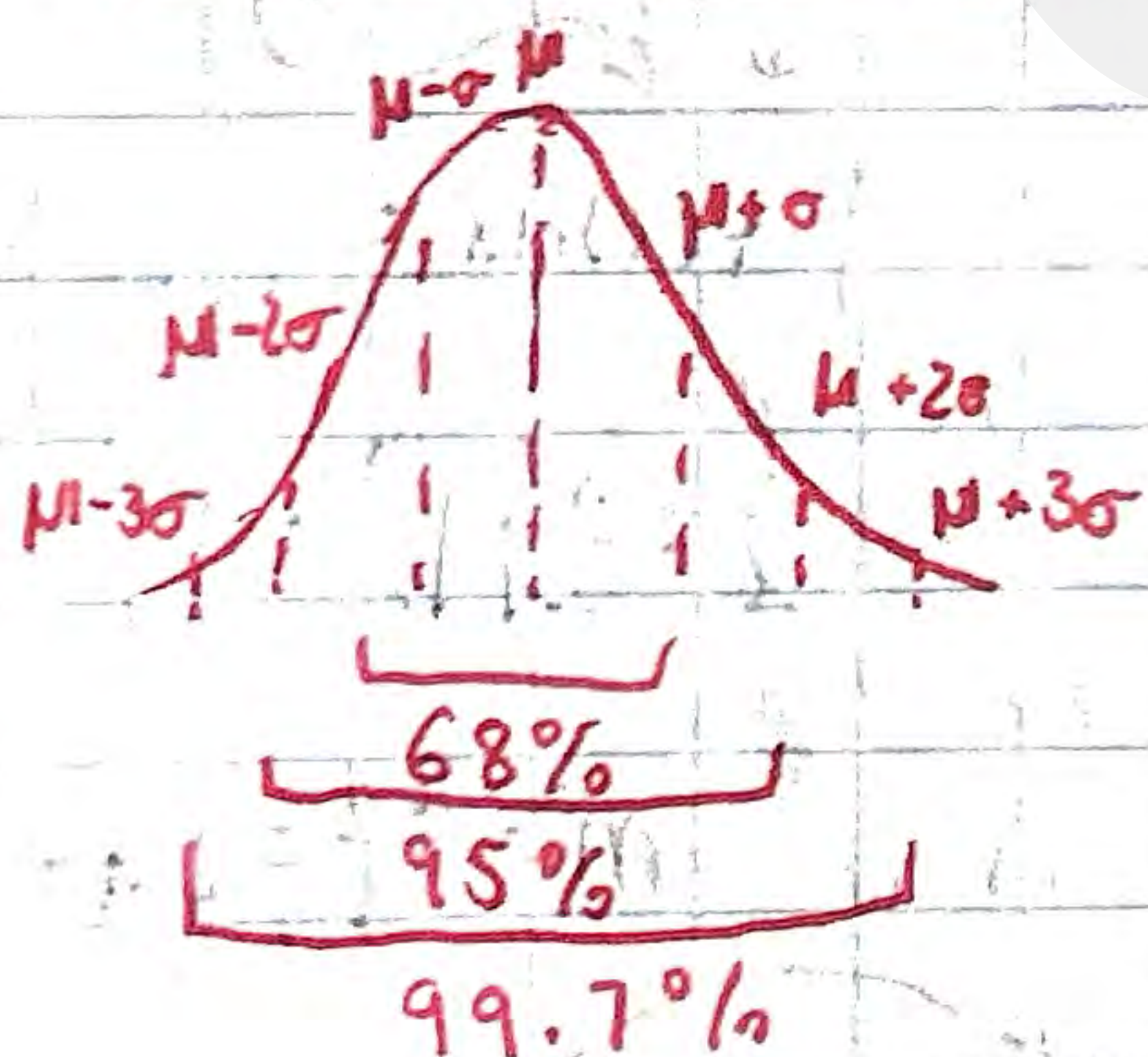
↓
Thus the curve is symmetrical

A random variable of normal distribution follows...

$$X \sim N(\mu, \sigma^2)$$

(The function itself is $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$)

Empirical rule: 68% of data within one standard deviation away from the mean



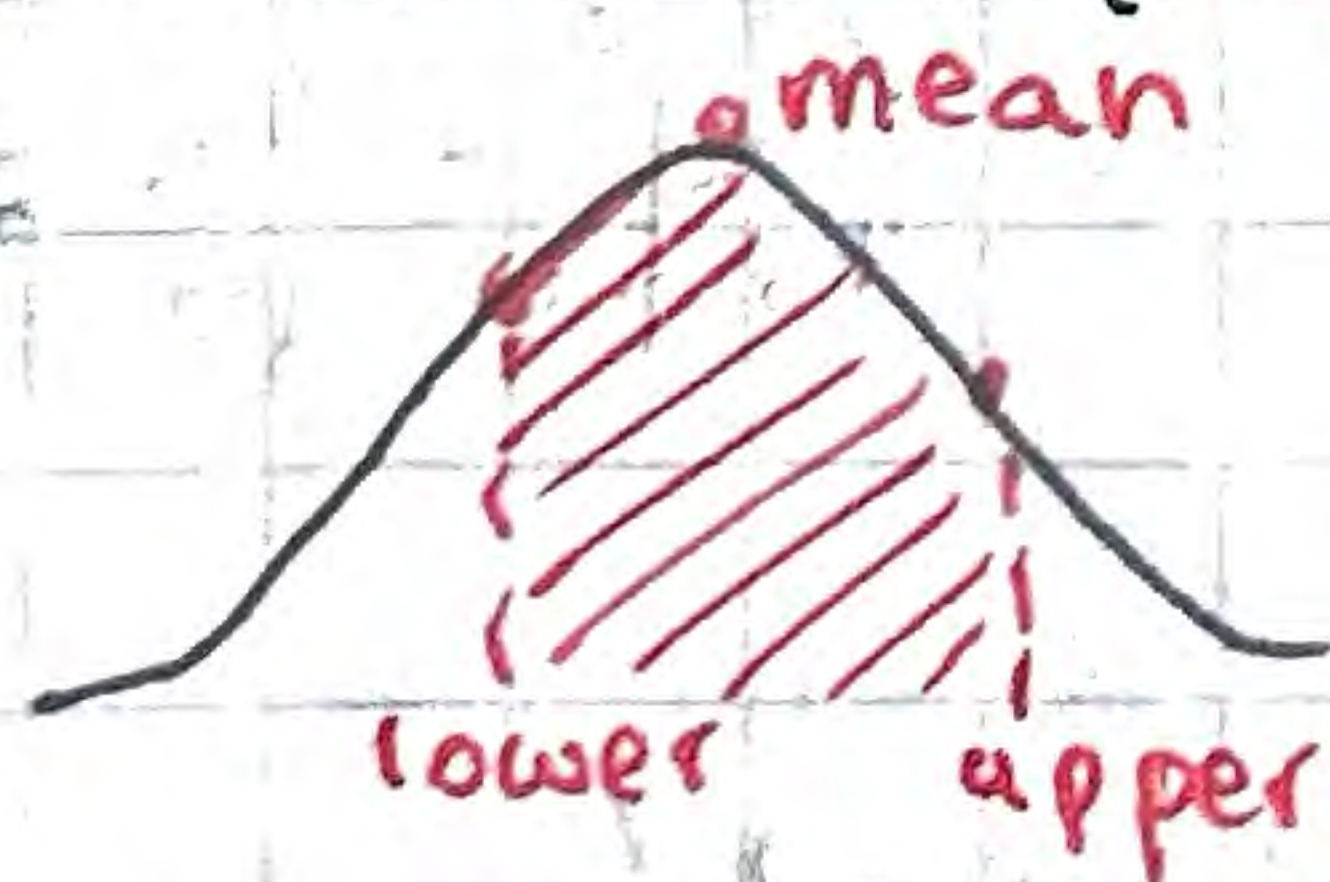
95% of data within two σ

99.7% of data within three σ

Graphing Calculator

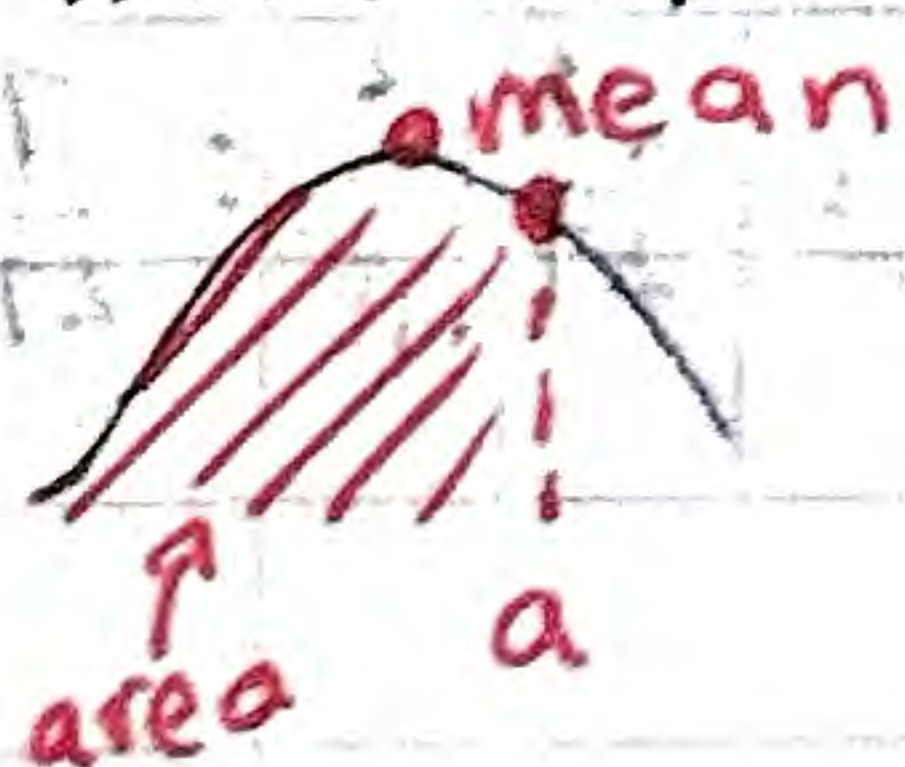
found at menu > probability > Distributions

norm Cdf (lower bound, upper bound, mean, standard deviation)



Calculates the probability over a certain range (the bounds can be $\pm\infty$)

inv Norm (area, mean, standard deviation)



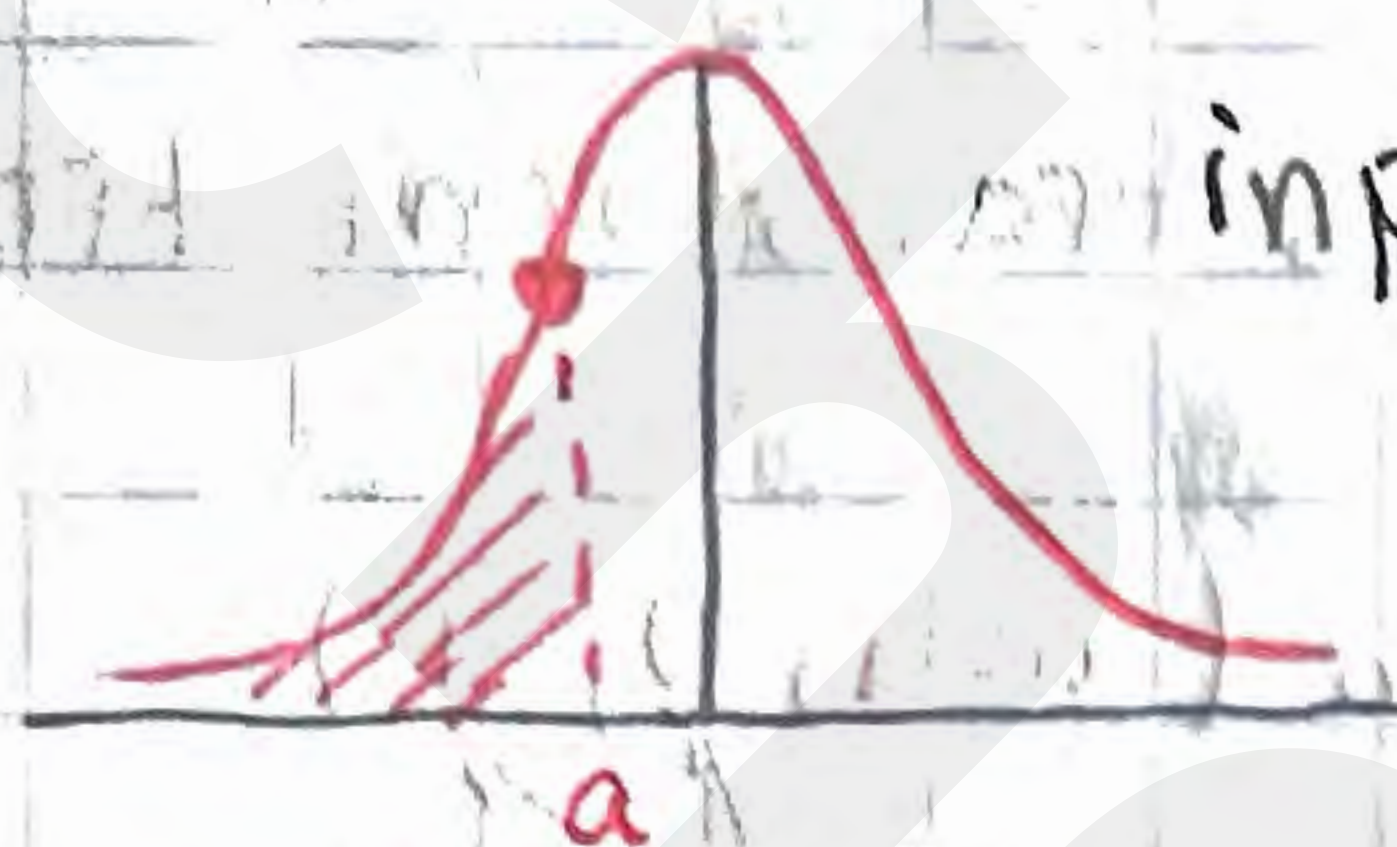
Calculates the upper bound for a certain probability

(lower bound will always be $-\infty$) make sure to input the area left of the 'a' you want to find

Inverse Normal

The area is the area left of the output value you want to find

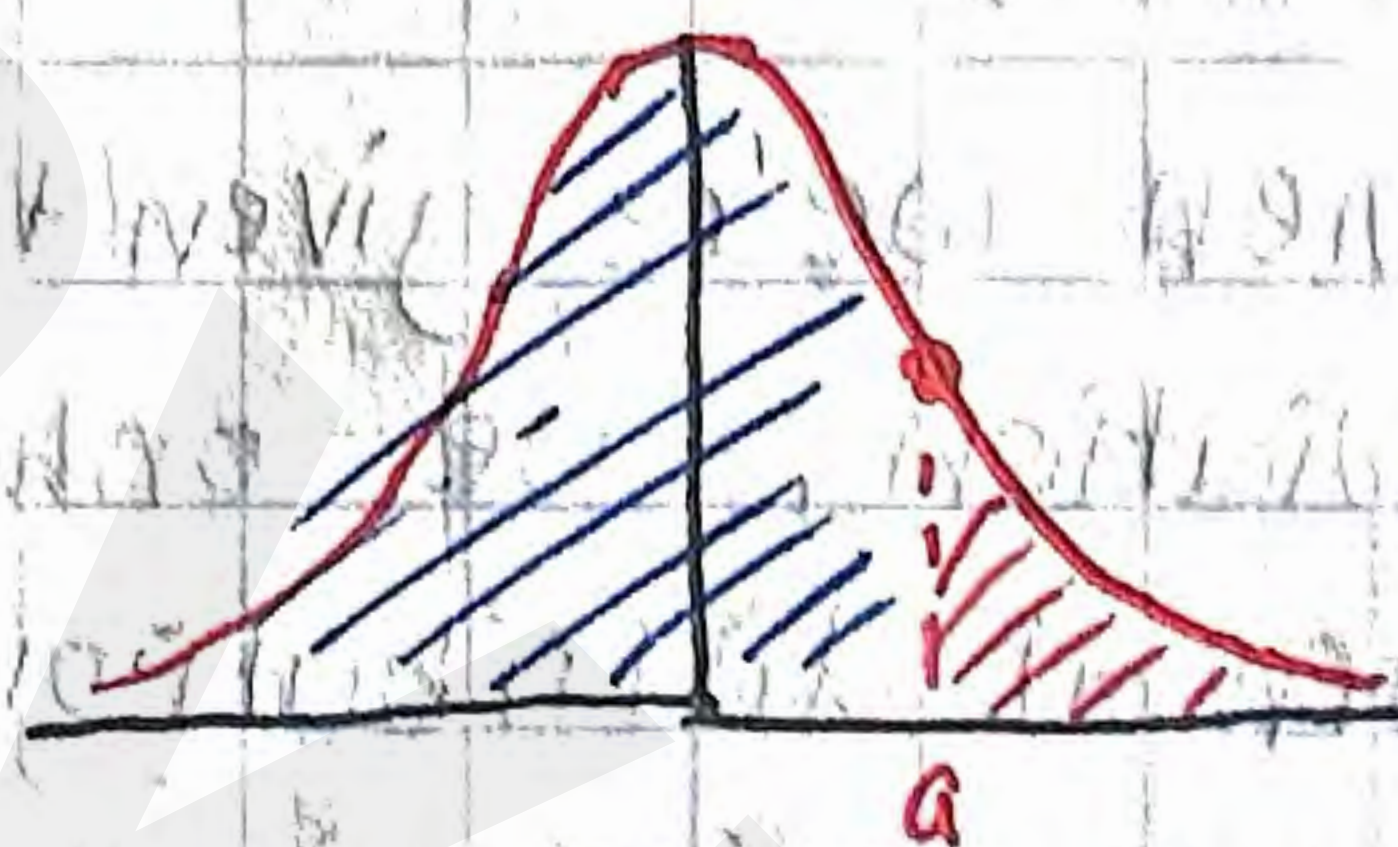
Thus $P(X < a)$ find $a \dots$



input area is just $P(X < a)$

However, when given $P(X > a)$ find $a \dots$

then input area is
 $1 - P(X > a)$

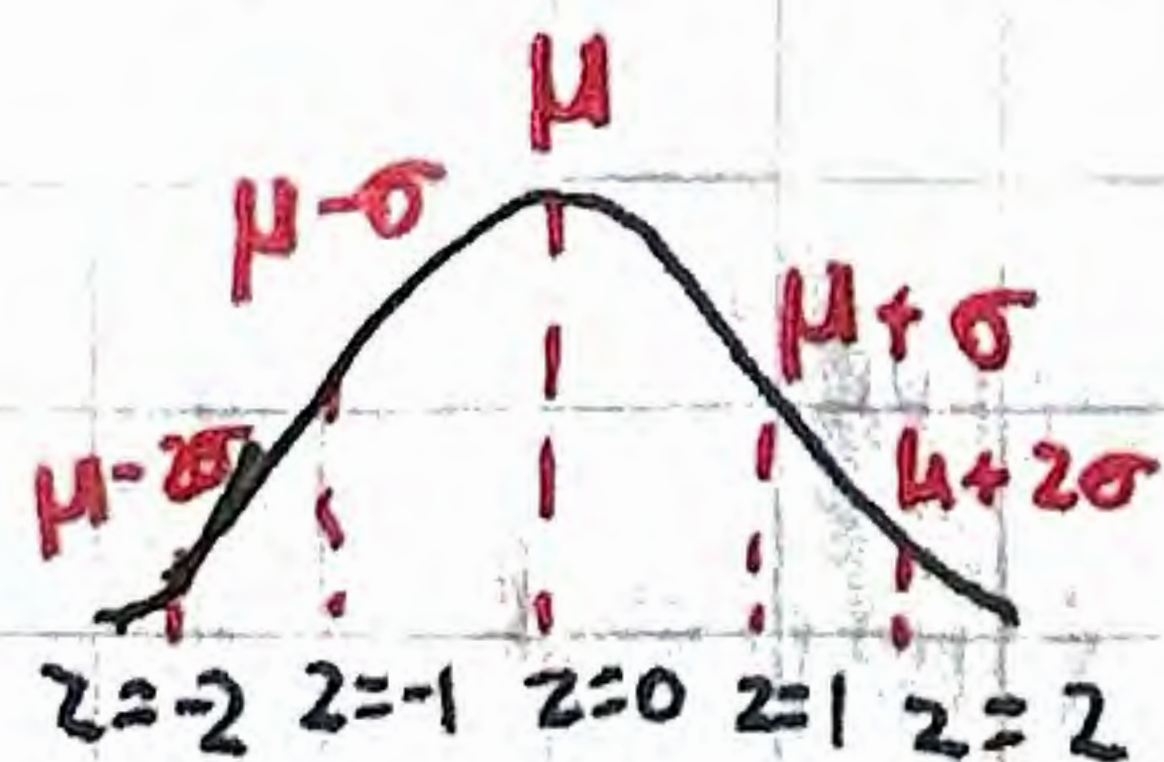


But what if we're asked to find the mean and standard deviation?

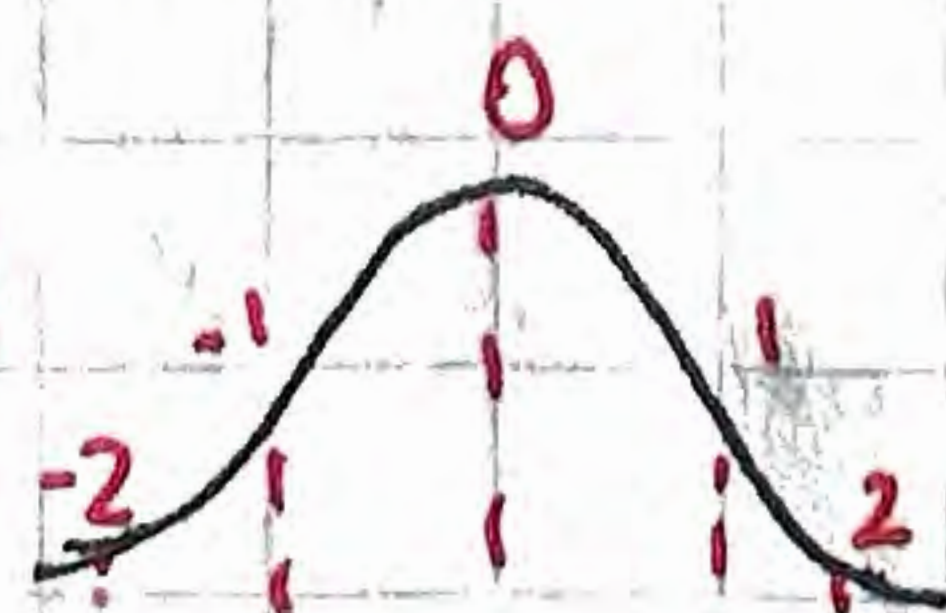
We utilize what is known as the Standard Normal Curve (fixed $\mu = 0$, $\sigma^2 = 1$) $Z \sim N(0, 1)$

• Normal curves are all the same, just transformed

> In a given normal curve, the corresponding values in a standard normal curve are known as the "Z-score"



and



↑
how many standard deviations from the mean

$$Z = \frac{X - \mu}{\sigma}$$

← shift mean to 0
↑ divide by σ to get 1, 2 etc.

Finding unknown mean and standard deviation

When you transform a normal curve, the probability bounded by a certain z-score will stay the same

- We can in turn utilize this transformation...

$$* P\left(Z \geq \frac{x - \mu}{\sigma}\right) = \text{inv Norm}(\text{area}, 0, 1)$$

values of the standard normal curve since we transformed the curve

- When we're given the mean or the standard deviation, we only need one other point from the normal distribution
- When we're given neither, we need two points from the normal distribution
 - We can use system of equations to solve for σ and μ

Vectors

Vectors - Quantities with both magnitude and direction

Notations $\begin{pmatrix} x \\ y \end{pmatrix} = \underbrace{\quad}_{x} \underbrace{\quad}_{y} = \langle x, y \rangle = x\hat{i} + y\hat{j}$

vectors do not have a "starting point", not fixed on plane
→ could also be $\overrightarrow{AB}$ to denote vector between points A, B

For higher dimensions..

$\begin{pmatrix} x \\ y \\ z \\ \text{etc.} \end{pmatrix} = \langle x, y, z, \text{etc.} \rangle = x\hat{i} + y\hat{j} + z\hat{k} + \dots$
"1" meaning unit

A unit vector is a vector of magnitude 1, denoted by $\hat{u}$

Magnitude - Length of a vector

Using pythagorean theorem for each component

$|\vec{v}| = \sqrt{x^2 + y^2 + z^2 + \dots}$ absolute value denotes magnitude

✓ vectors of the same direction can be factorized into the other. e.g. $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$

Direction

Relationships between vectors

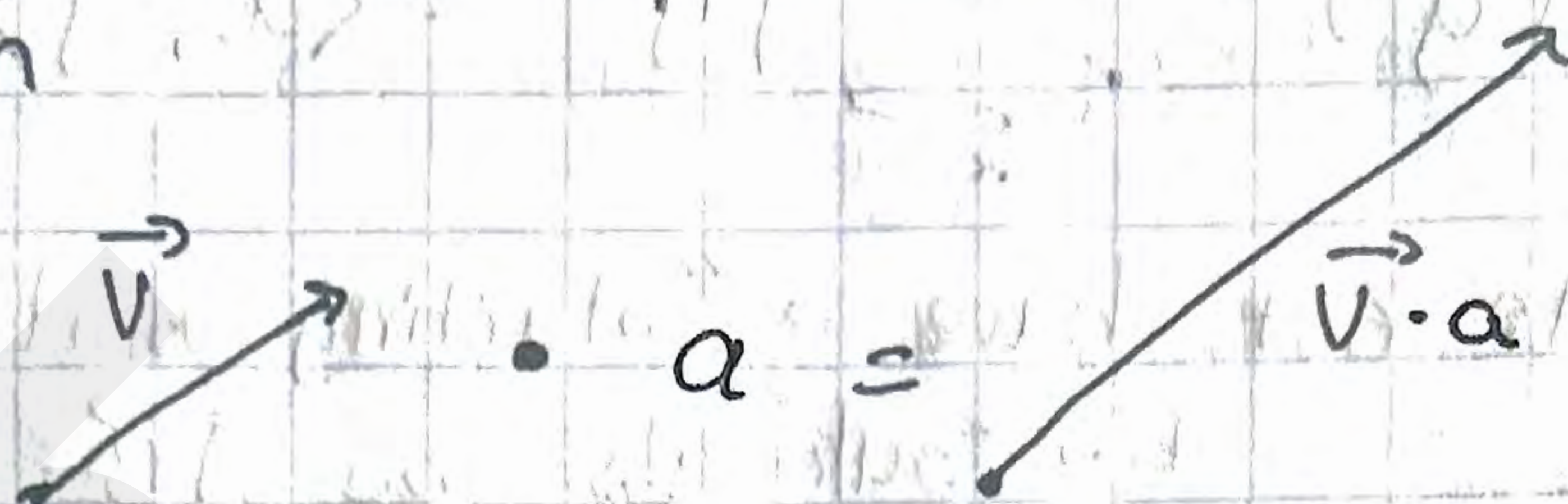
- Equal vectors have same magnitude and direction
- Negative vectors have same magnitude, negative direction
- Parallel vectors have same direction, any magnitude

starting point doesn't matter



Vector - Scalar multiplication

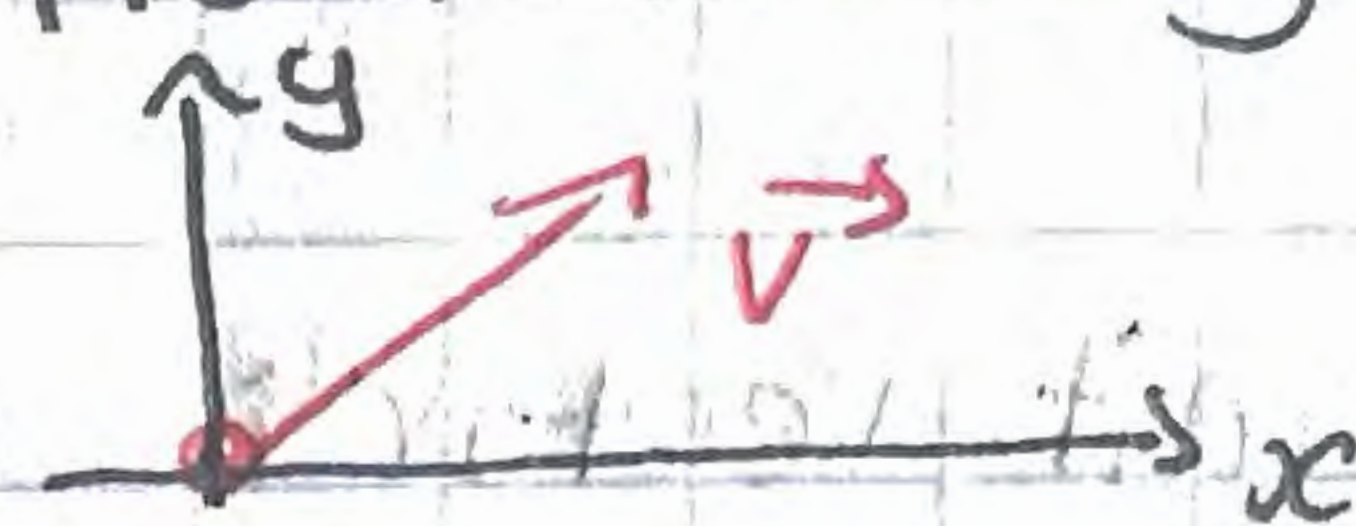
Multiplying a vector by a scalar does not change direction



Magnitude is multiplied by the scalar quantity

Position vector - A vector starting from the origin

• Denoted by $\vec{OA}$ *origin*



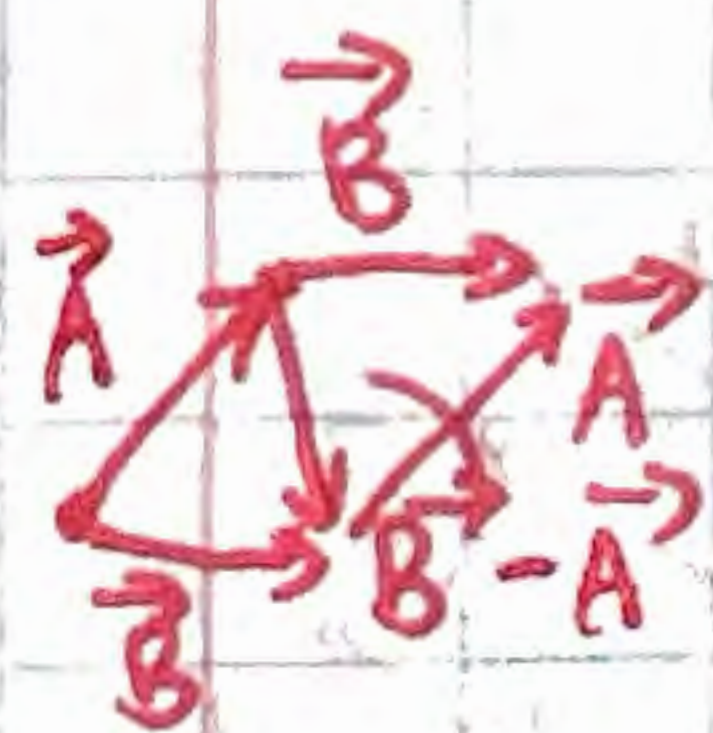
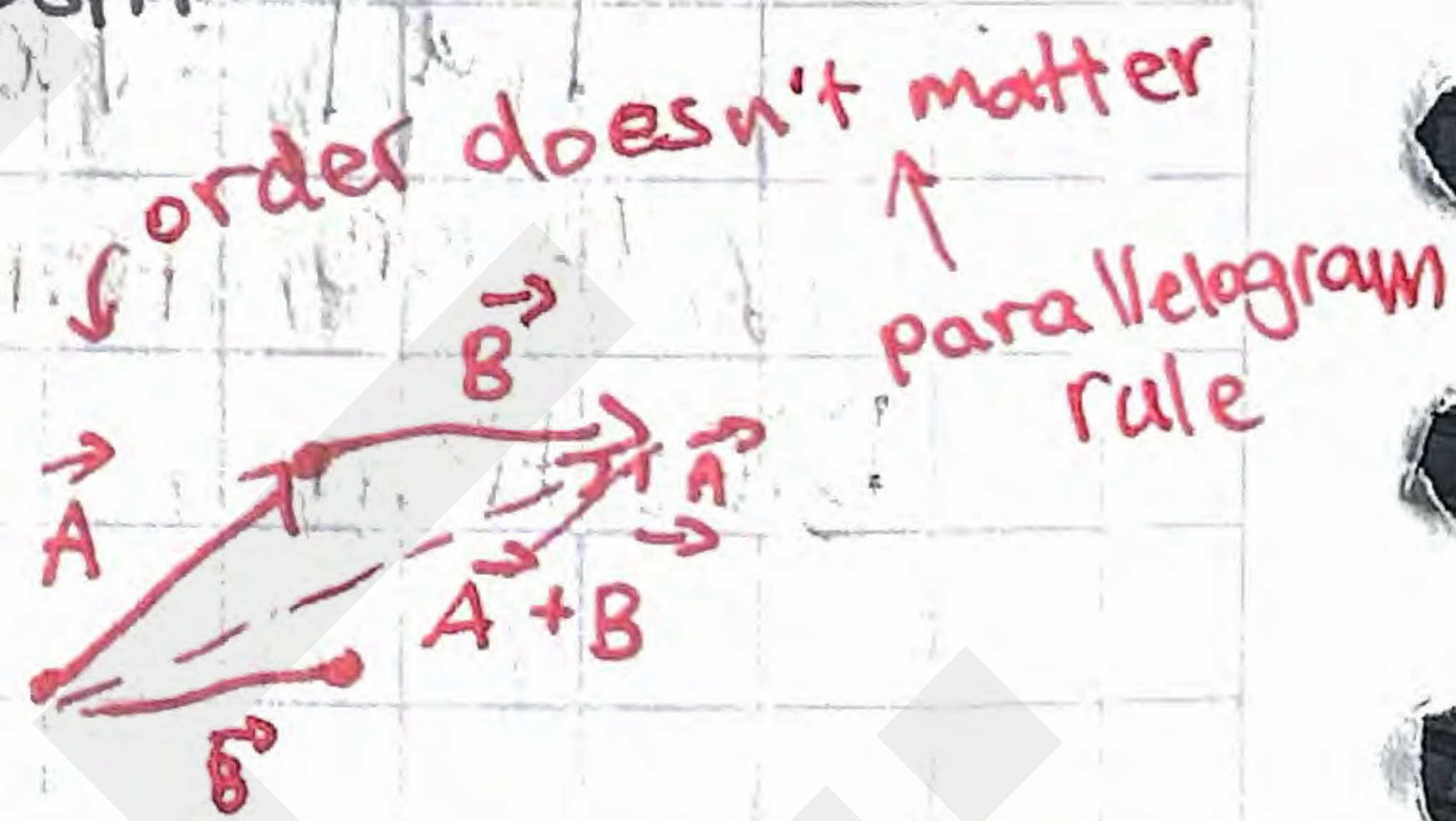
Resultant vector - Vectors that measure change rather than tracking points in space

• Defined by terminal position - initial position

Addition and subtraction of vectors

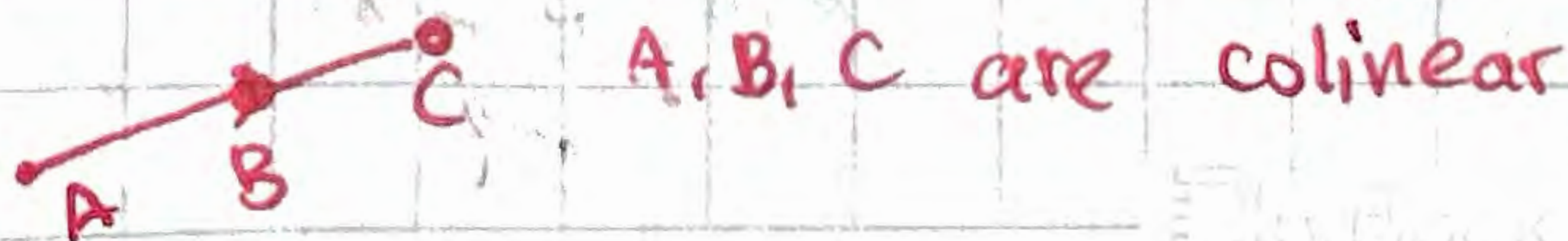
• An addition of vectors involve placing vectors for the tail of one corresponds to the head of another

• The resultant vector from addition is the vector from the tail of the first to the tail of the last



A subtraction is the same as addition with negative terms where the vector's head-tail is flipped

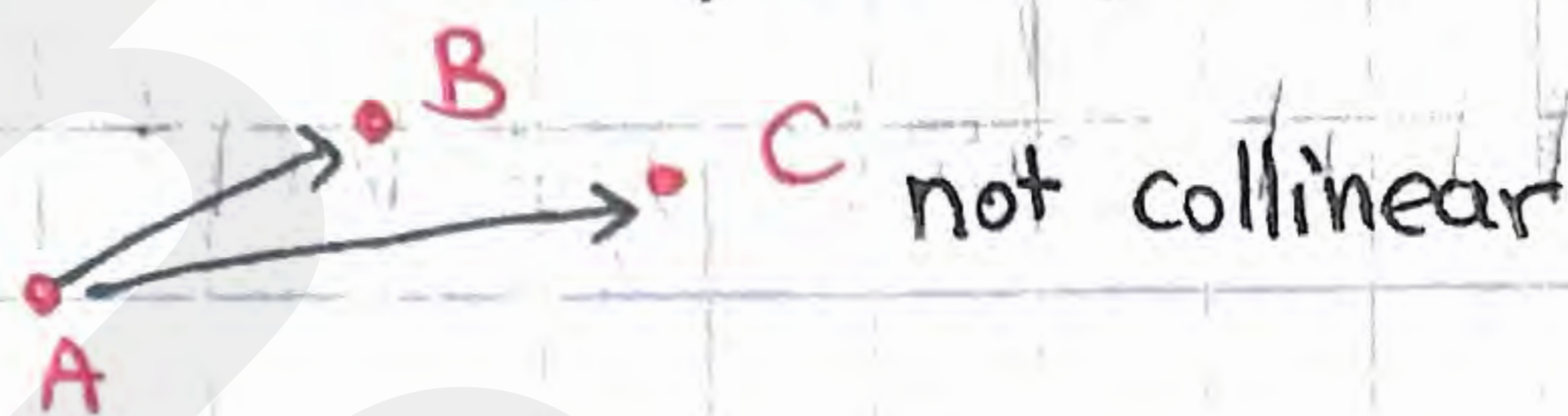
Collinearity - Points that together form a single straight line



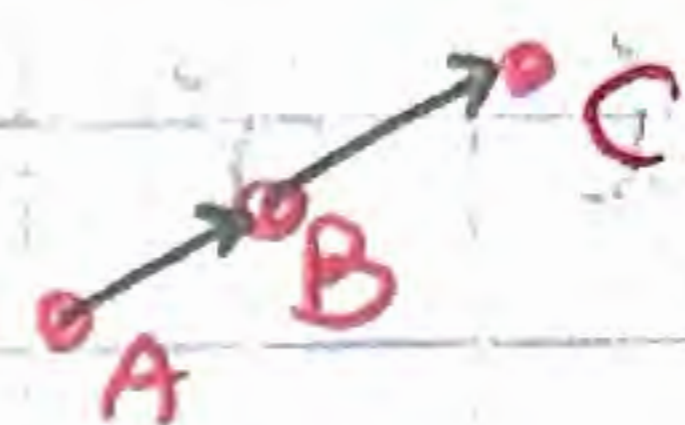
Vectors and point collinearity

Since the points must form a single straight line, we can also use vectors to justify collinearity.

- Points are collinear when vectors starting from first point to other points are all in the same direction



not collinear



collinear

Unit Vector

You can scale a vector into a unit vector (magnitude one)

let $\vec{u} = \begin{pmatrix} a \\ b \end{pmatrix}$ then $\hat{u} = \begin{pmatrix} a_0 \\ b_0 \end{pmatrix}$ must satisfy $a_0^2 + b_0^2 = 1$

You can find a_0 as follows...

$$a = a_0 \cdot |\vec{u}|, \quad b = b_0 \cdot |\vec{u}|$$

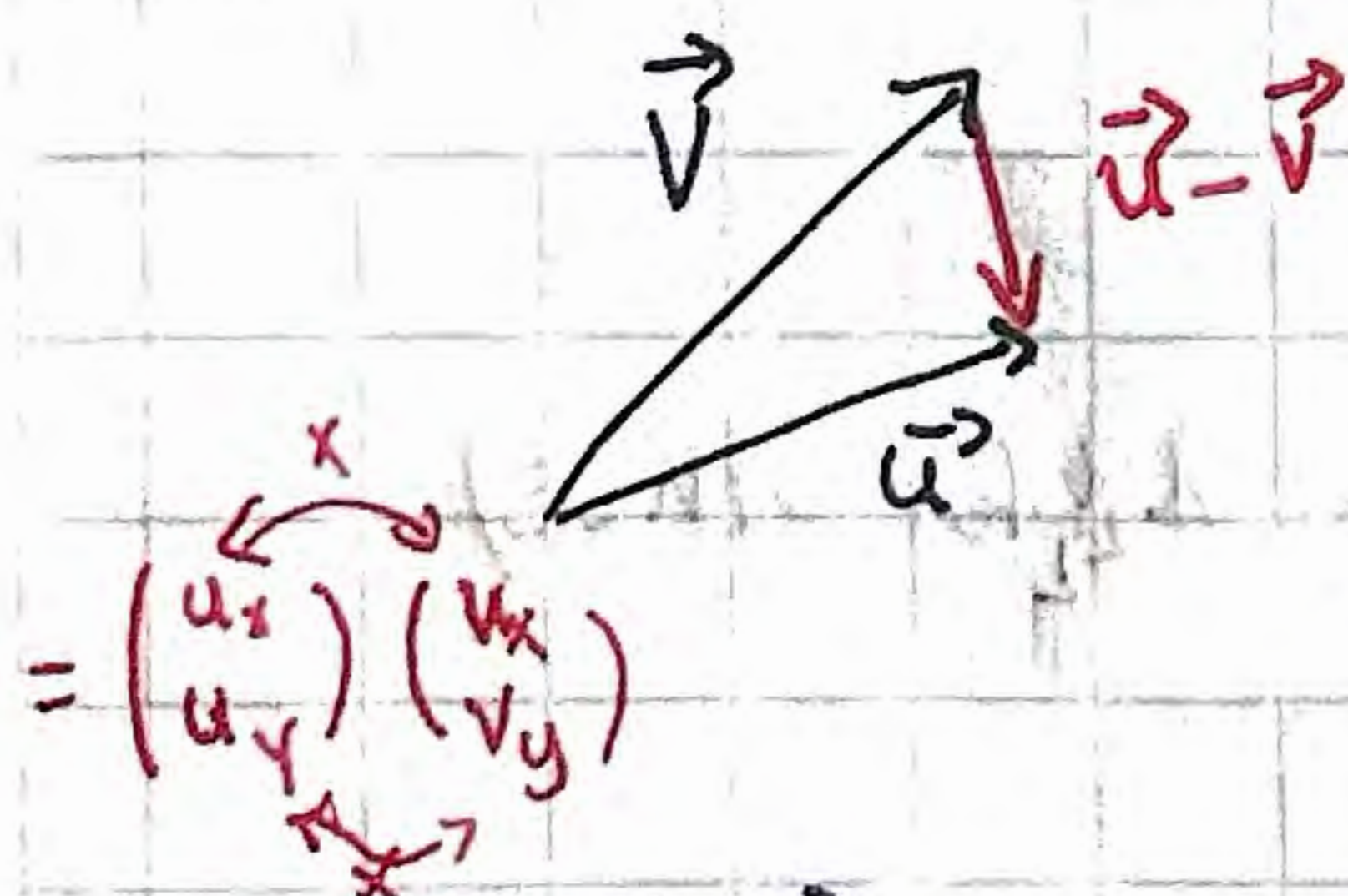
In other words, $\hat{u}$ is just $\vec{u}$ divided by its magnitude $|\vec{u}|$

Dot / Scalar Product

Product between vectors such that its result always yields a scalar

$$\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$$

angle
between
vectors



$$|\vec{u} - \vec{v}|^2 = |\vec{u}|^2 + |\vec{v}|^2 - 2|\vec{u}||\vec{v}|\cos\theta$$

$$(\vec{u} - \vec{v}) \cdot (\vec{u} - \vec{v}) = |\vec{u}|^2 + |\vec{v}|^2 - 2|\vec{u}||\vec{v}|\cos\theta$$

$$\vec{u} \cdot \vec{u} - 2\vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{v} = \vec{u} \cdot \vec{u} + \vec{v} \cdot \vec{v} - 2|\vec{u}||\vec{v}|\cos\theta$$

$$\vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}|\cos\theta$$

dot product can be used
to find
angle between
the vectors

$$\vec{u} \cdot \vec{v} \text{ is also equal to } (u_x)(v_x) + (u_y)(v_y)$$

Dot / Scalar Product

Properties when two vectors form a dot product of zero
 $\cdot \cos \theta$ is zero, thus the two vectors are perpendicular

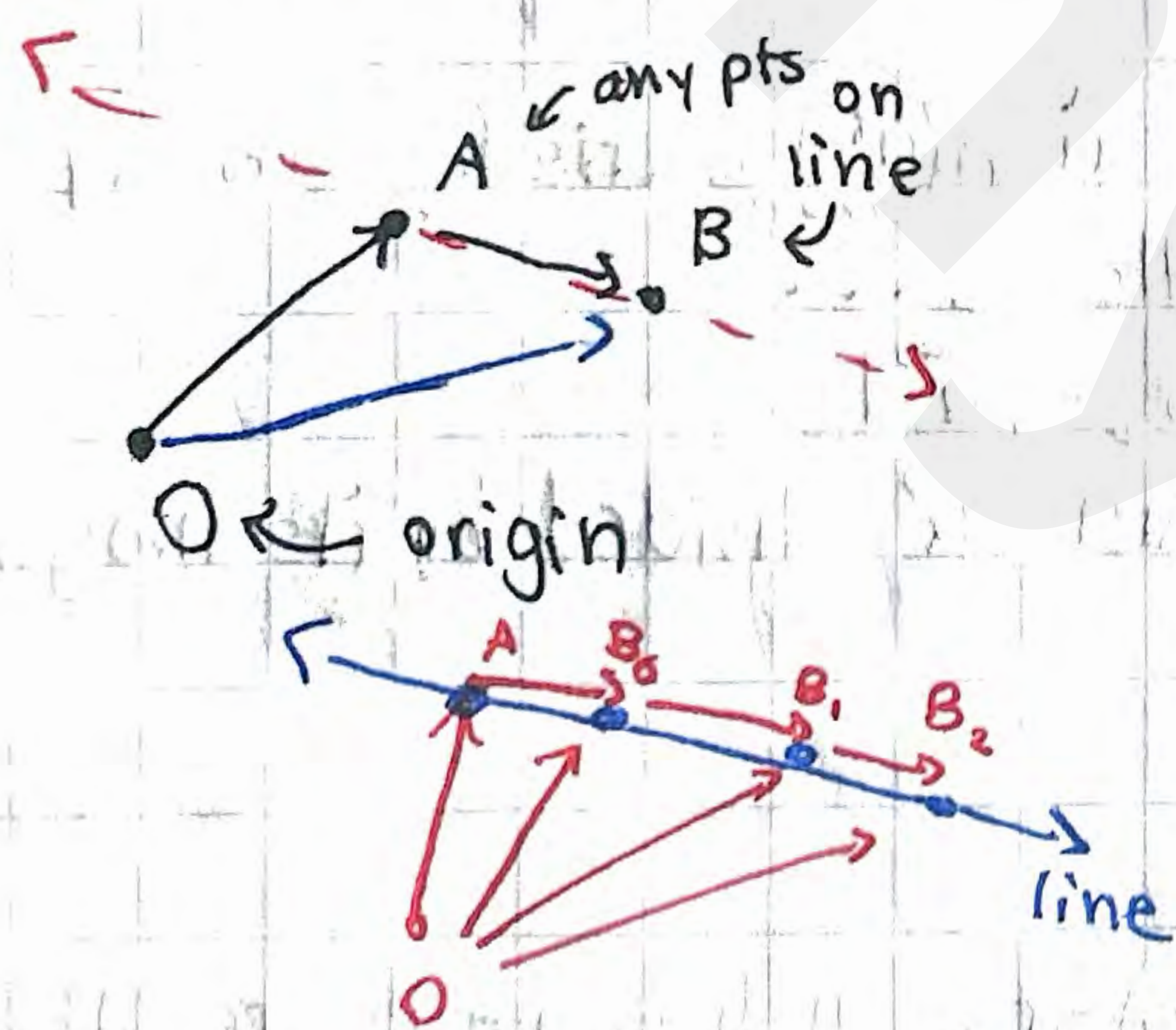
A vector dot itself is just equal to its magnitude squared

$$> \vec{a} \cdot \vec{a} = |\vec{a}| |\vec{a}| \cos(0) = |\vec{a}|^2$$

Vector equation of a line

A line can be defined by a vector equation

(where the vector points to every point on the line)



$\vec{OB}$ points to a point on the line.

If you have infinite vectors pointing to all pts that compose a line, then that'll provide the equation of the line

Thus...

$$l = \vec{OA} + \lambda \vec{AB}$$

↑
anchors
line to
coordinate
plane

↑
direction
of line

(l and λ are unique to each line so $\lambda_1 \neq \lambda_2$ for others)

Vector equation of lines

We can thus obtain the intersection of vector equation of lines

- We do this by solving for λ s when x is the same, then when y is the same, etc. (and verifying if the unknown λ s are the same)

↳ if λ s are not the same, then they will not intersect

> The angle at which they intersect can be found with the dot product's $\cos \theta$ term

↳ Since only direction determines θ , using only the directional vector $\vec{AB}$ for dot product is sufficient

E.g. $r_1 = \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} + s \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$
 $r_2 = \begin{pmatrix} 6 \\ 2 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 4 \\ 8 \end{pmatrix}$

analyze by components > $x: 3 + s = 6 + t(0)$

↳ $s = 3$

$y: 0 + s = 2 + 4t$

↳ $t = \frac{1}{4}$

$z: -1 + s = 0 + 8t$

$-1 + 3 = 8(\frac{1}{4})$

$2 = 2 \checkmark$

Yes, (x, y, z)

exists for $r_1 = r_2$

} we got s, t already, but does it satisfy z ?

Dot Product Properties

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}, \text{ product, order doesn't matter}$$

$$\vec{a} \cdot \vec{a} = |\vec{a}|^2, \text{ by pythagorean theorem/definition of dot product}$$

Distributive
Property works

$$(\vec{a} + \vec{b}) \cdot (\vec{c} + \vec{d}) = \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c} + \vec{a} \cdot \vec{d} + \vec{b} \cdot \vec{d}$$

Cross product

$$\vec{u} \times \vec{v} = \begin{pmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{pmatrix}$$

comes from

$$\vec{u} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} \quad \vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$\text{e.g. } \vec{u} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} \quad \vec{v} = \begin{pmatrix} -3 \\ 4 \\ 1 \end{pmatrix}$$

$$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ -3 & 4 & 1 \end{vmatrix}$$

parts of the
matrix not same
row/column

$$= \hat{i} \begin{vmatrix} 2 & -3 \\ 4 & 1 \end{vmatrix} - \hat{j} \begin{vmatrix} 1 & -3 \\ -3 & 1 \end{vmatrix} + \hat{k} \begin{vmatrix} 1 & 2 \\ -3 & 4 \end{vmatrix}$$

Cross such that $\rightarrow$ alternate
signs

$$2 \cdot 1 - (-3 \cdot 4) = 14$$

$$= 14\hat{i} + 8\hat{j} - 10\hat{k}$$

$$\hat{i} - \hat{j} + \hat{k} \dots$$

Makesure
Signs alternate

The cross goes from

$$\begin{bmatrix} u_1 & u_2 \\ v_1 & v_2 \end{bmatrix} = u_1 v_2 - u_2 v_1$$

what does the cross product represent?

• It obtains a vector perpendicular to both vectors



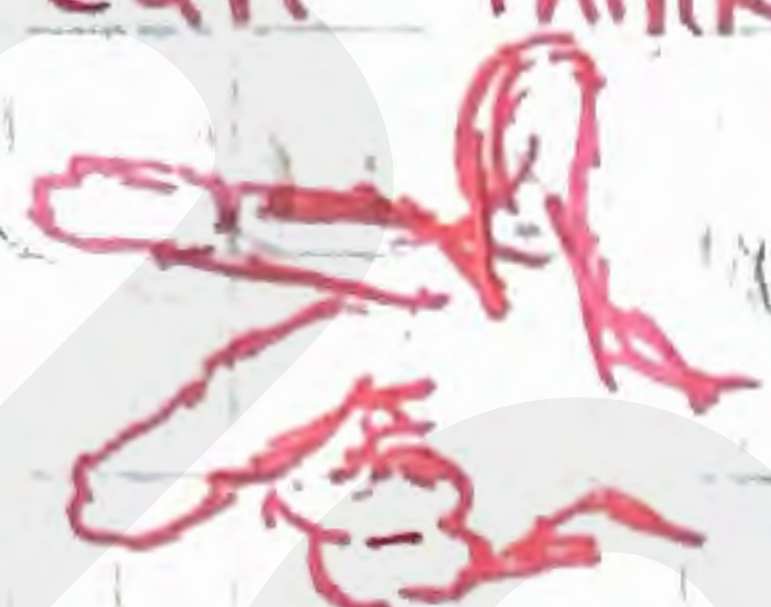
Thus, parallel have infinite
perpendicular vectors (cross product yields 0)

Cross Product

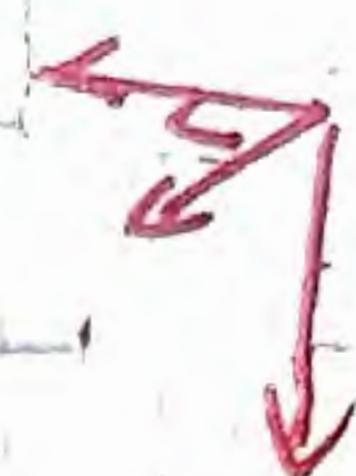
Since cross products are highly sign specific

$$\vec{u} \times \vec{v} = -\vec{v} \times \vec{u} \quad \Leftarrow \text{Thus order does matter}$$

You can think of it as the right hand rule



where opposite $\vec{u}$ and $\vec{v}$ is

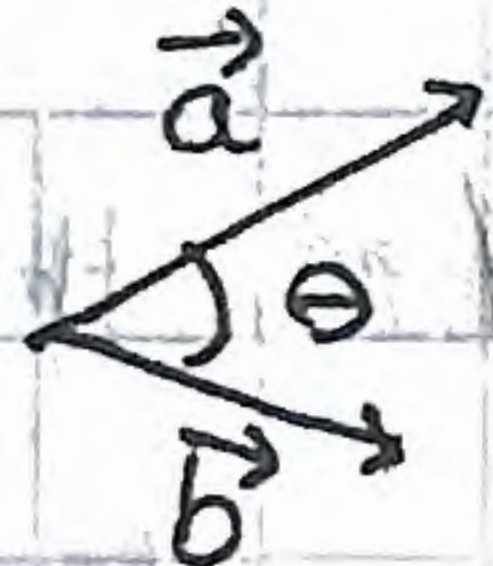


The magnitude of a cross product is...

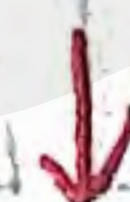
$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$$



where

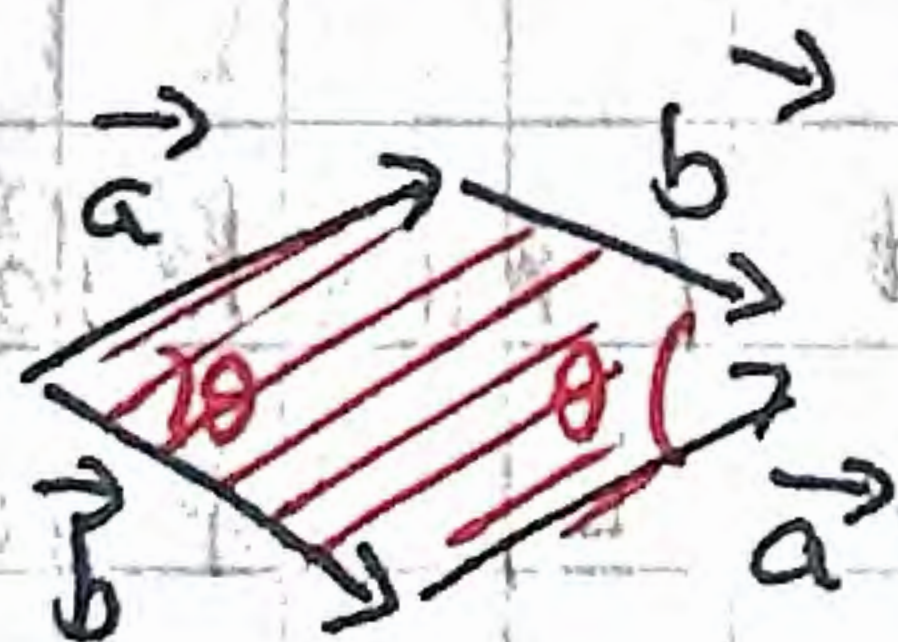


look similar? (and cross product looks like shoelace theorem)



$$A = \frac{1}{2} ab \sin \theta$$

Thus, the magnitude is... $\hookrightarrow = \frac{1}{2} |\vec{a} \times \vec{b}|$



or the area of the parallelogram

Other properties...

$$\vec{a} \cdot (\vec{a} \times \vec{b}) = \vec{b} \cdot (\vec{a} \times \vec{b}) = 0$$

$\hookrightarrow$ since the cross is perpendicular

$\hookrightarrow$ Thus dot = 0

Vector line applications

Recall

$$\ell = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} + t \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

Parametric form (with respect to t)

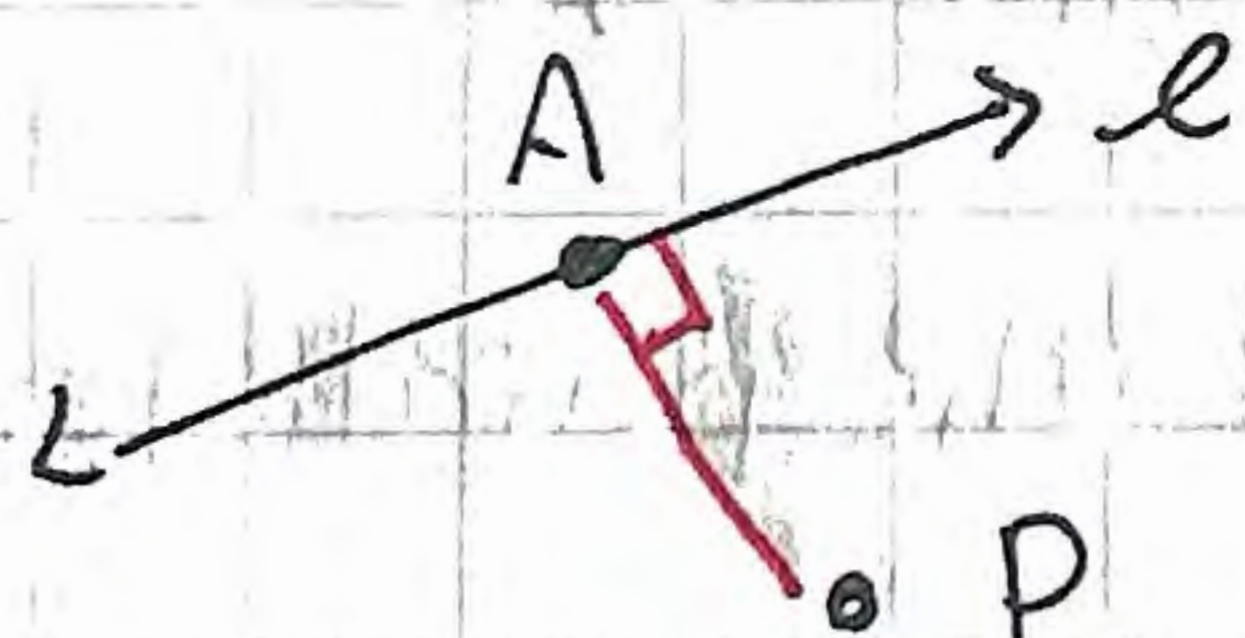
$$x = u_1 + v_1 t, \quad y = u_2 + v_2 t, \quad z = u_3 + v_3 t$$

Cartesian form (with respect to x, y, z)

$$t = \frac{x - u_1}{v_1}, \quad t = \frac{y - u_2}{v_2}, \quad t = \frac{z - u_3}{v_3}$$

thus...
$$\frac{x - u_1}{v_1} = \frac{y - u_2}{v_2} = \frac{z - u_3}{v_3}$$

Given line, we can find point A on the line closest to another point P



The closest point A must form $\vec{AP} \perp \ell$

• For them to be $\perp$, dot product must be zero

↳ Thus $\ell \cdot \vec{AP} = 0$

↳ solve for parametric value to get $\vec{OA}$

How about between two lines?

- Just like points, the shortest vector is perpendicular to both line vectors
- This is the cross product!

Shortest vector between two lines

- Let point A be on ℓ_1 , B on ℓ_2
- Thus $\vec{AB}$ perpendicular to both ℓ_1 and ℓ_2
- As such $\vec{AB}$ parallel to $\ell_1 \times \ell_2$
- Since we're using direction, we use the directional vectors of ℓ_1 and ℓ_2 and cross them.

↳ They're parallel, meaning $k(\ell_1 \times \ell_2) = \vec{AB}$ (scaled)

E.g. $\ell_1: x=t, y=1-t, z=2+t$
 $\ell_2: x=3-s, y=-1+2s, z=4-s$

$$\therefore \ell_1 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \quad \ell_2 = \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} + s \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

let $A = (t, 1-t, 2+t)$, $B = (3-s, -1+2s, 4-s)$

$$\vec{AB} = \begin{pmatrix} 3-s-t \\ -2+2s+t \\ 2-s-t \end{pmatrix} \quad \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Thus $\begin{pmatrix} 3-s-t \\ -2+2s+t \\ 2-s-t \end{pmatrix} = k \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$, solve 3 eq. 3 variable system

After getting k , you can get $|\vec{AB}|$ through $|k \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}|$

• or s, t if asking for A, B

Vector equation of planes

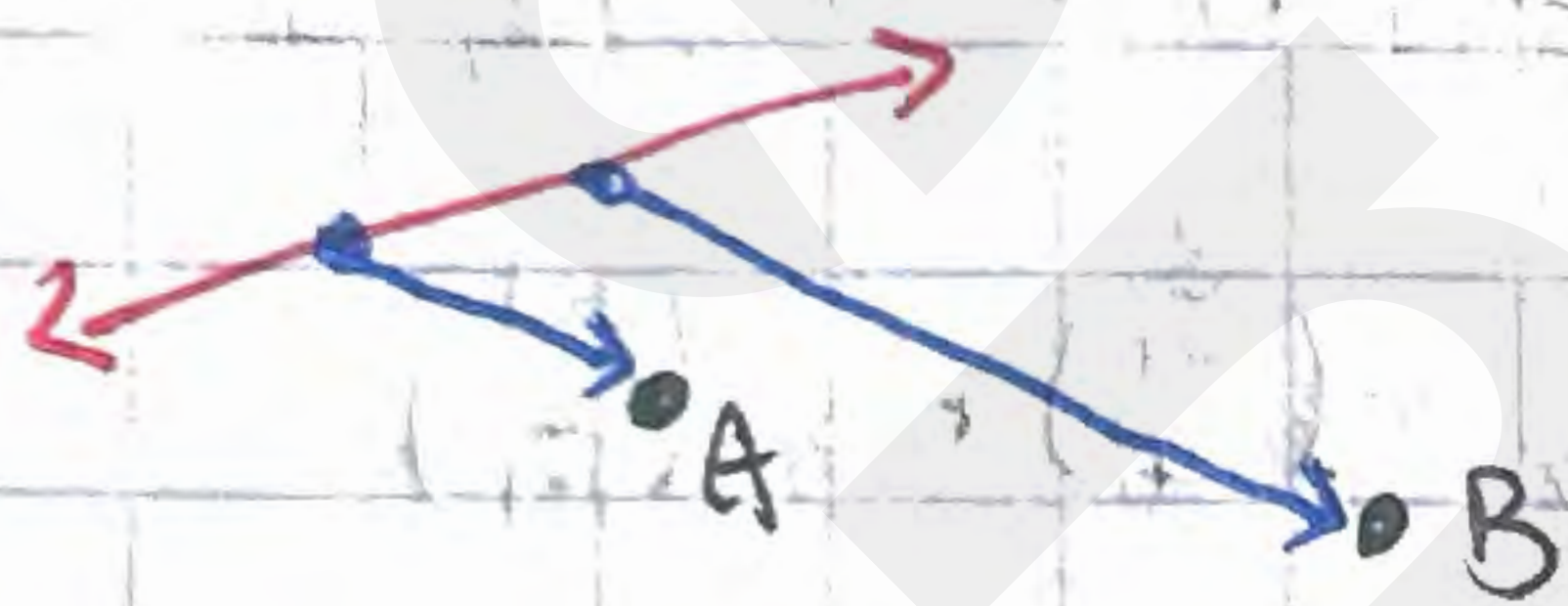
Any point on a plane can be found from the sum of two vectors (scaled)

↳ From vector equation of a line

$$l: p + \alpha c$$

all pts on a line

by adding a scaled directional vector, any vector can be obtained



Thus any point can be formed from 2 vector

$$\pi: p + \alpha \vec{c} + \beta \vec{d}$$

plane

Point on the plane

directional vectors

parameters

Parametric form

$$\pi: \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix} + \alpha \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} + \beta \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

Thus

$$\begin{aligned} x &= p_1 + \alpha u_1 + \beta v_1 \\ y &= p_2 + \alpha u_2 + \beta v_2 \\ z &= p_3 + \alpha u_3 + \beta v_3 \end{aligned}$$

parametric of α, β parameters

Cartesian form would be getting rid of α, β

Vector equation of planes

Cartesian form

Method 1

↳ Eliminate α , get two equations

↳ Solve for β , equate, simplify

$$\dots ax+by+cz=d$$

E.g. $\pi: \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix} + \beta \begin{pmatrix} 5 \\ 1 \\ 3 \end{pmatrix}$

$$\begin{cases} x = 2 - \alpha + 5\beta \\ y = 3 + 4\alpha + \beta \\ z = 4 + 3\alpha + \beta \end{cases} \Rightarrow \begin{cases} 3x - 2 = 2 + 14\beta \\ 4x - y = 5 + 17\beta \end{cases}$$

$$\therefore \frac{3x-2-2}{14} = \frac{4x-y-5}{17}$$

$$-5x + 14y - 17z = -36$$

Method 2

↳ Cross product

• By crossing two directional vectors, you get $ax+by+cz$

• To get d , you "fix" the direction at a point by plugging in the known point (i.e. p , without coeff.)

E.g. $\left(\frac{1}{3}\right) \times \left(\frac{5}{3}\right) = -5x + 14y - 17z$

We know $\left(\frac{2}{3}\right)$ so $-5(2) + 14(3) - 17(4) = -36$

→ This is also the equation of $\perp$ vector (w/o x, y, z)

Plane-line intersection

• An intersection means equate, plug-in

• When given vector equation of line (parametric), plug in $x(\lambda)$, $y(\lambda)$, $z(\lambda)$ and solve for λ

E.g.

$$l: \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ -6 \end{pmatrix}$$

$$\pi: x - 2y + 3z = 26$$

$$\hookrightarrow x = -1 + 3\lambda$$

$$y = 2 - 2\lambda$$

$$z = 3 - 6\lambda$$

plug in and solve

can be given by $\cos(90^\circ - \theta)$

→ Finding angle of intersection

↳ Do cross product between normal and line

Perpendicular vector to plane

Recall the coefficients of a plane equation actually come from cross product (perpendicular vector to the two directional vectors of plane)



Thus the equation

$$ax + by + cz = d$$

indicates vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is perpendicular to plane

Why is this useful?

Shortest distance to point

↳ $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is direction, the point (fixed part) is the given point

↳ After obtaining equation of line, then find intersection with plane

↳ Thus, a vector between the two points, find magnitude

E.g. $A(2, -1, 3)$

$$\pi: x - y + 2z = 27$$

$$l = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

$$\begin{aligned} x &= 2 + \lambda \\ y &= -1 - \lambda \\ z &= 3 + 2\lambda \end{aligned} \rightarrow \begin{aligned} 2 + \lambda - (-1 - \lambda) + 2(3 + 2\lambda) &= 27 \\ 2 + \lambda + 1 + \lambda + 6 + 4\lambda &= 27 \\ 6\lambda + 9 &= 27 \end{aligned}$$

$$P: x = 5, y = -4, z = 9$$

$$6\lambda = 18$$

$$A(2, -1, 3)$$

$$\leftarrow \lambda = 3$$

$$AP = \begin{pmatrix} 3 \\ -3 \\ 6 \end{pmatrix}$$

$$|AP| = \sqrt{54}$$

Angles of intersection

• Between planes

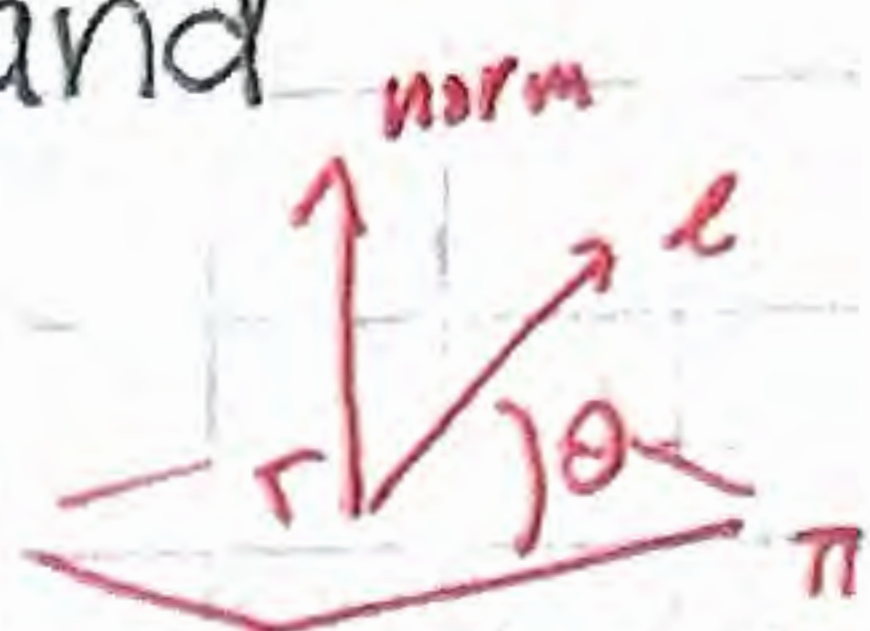
↳ Same as $\angle$ between normal lines

• Use dot product

✓ Coefficient of plane equation

Angles of intersection

- Between plane and line
 - ↳ Find angle between normal of plane and line (dot product)
 - ↳ Then, subtract the angle from 90° ($\frac{\pi}{2}$)
 - ↳ The absolute value of the final angle is the angle of intersection
- Between lines
 - ↳ Dot product the two lines to get $\cos \theta$, thus obtaining θ



Intersections

Between lines

- Equate x, y, z (the parameter λ may be distinct for the two lines, thus treat them differently)
- If all 3 equations of x, y, z are satisfied with the same parameter values then the intersection is a point
 - For more than lines, then make sure same results are obtained when any two of them are compared
- ↳ If not, then no intersection

Between line and plane

- Plug in x, y, z with respect to the parameter into the Cartesian plane equation
- Find the point of the found x, y, z values
- They should intersect at a point
 - ↳ However, if all λ parameters cancel out and...
 - a) an impossible equation obtained no solution
 - b) an always true equation obtained line (∞) solution



Intersections

- Between two planes
- Use cartesian form and solve linear equations in term of one variable

$$A_1x + B_1y + C_1z = D_1$$

$$A_2x + B_2y + C_2z = D_2$$

E.g. obtain $x = f(y) = g(z) \leftarrow \text{parametric}$
Set $x = \lambda$ then solve for $y(\lambda), z(\lambda)$

• Hence, final: $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \lambda \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$

- However, if planes are distinct and parallel then no solution

- Intersection between three planes follow similarly from two plane intersection
↳ Except they typically intersect at a point instead of a line

* Can also be found with cross product

$$\begin{pmatrix} A_1 \\ B_1 \\ C_1 \end{pmatrix} \times \begin{pmatrix} A_2 \\ B_2 \\ C_2 \end{pmatrix} \text{ can yield direction of intersection}$$

- Then find any point in which two planes intersect (let one variable be zero and solve for others)
↳ Hence, direction + fixed point is line